

soils4med

SOIL health monitoring and information systems FOR
sustainable soil management in the MEDiterranean region

MedSIS manual *Taxonomies*



DATASETS



MAPS



DOCUMENTS



GEOSTORIES



DASHBOARDS

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1. Overview

A Soil Information System (SIS) is an integrated information system designed to facilitate the storage, analysis, management and dissemination of soil data.

To assess and monitor soil health and promote sustainable soil management in the Mediterranean region, there is a pressing need to improve the availability and accessibility of soil data.

One of the objectives of the SOILS4MED project (SOIL health monitoring and information systems FOR sustainable soil management in the MEDiterranean region) which addresses this need by designing and developing a decentralised network of country-based soil information systems.

This network aims to improve the interoperability, management, governance, and utilization of soil data across the Mediterranean region by establishing a federated architecture of SISs, starting with the development of a harmonized data model aligned with the World Reference Base for Soil Resources (WRB 2022), alongside the implementation of core functionalities for the ingestion, integration, and management of both legacy and newly acquired soil data.

The SIS is essentially a  of resources related to soil data. It is based on Geonode (<https://geonode.org/>), whose resources include datasets, maps, documents, geostories and dashboards. In addition to the standard catalogue functionalities for managing , MED-SIS includes an additional tool, known as the .

The system uses several taxonomies to describe and manage data. The main source of these taxonomies is the World Reference Base for Soil Resources ([WRB 2022](#)) standard

In MED-SIS, it is possible to access the taxonomy via two endpoints in JSON format:

- "Taxonomies": `.../api/backoffice/taxonomies/`
- "taxonomy-values": `.../api/backoffice/taxonomy-values/`



Figure 1 - MED-SIS API - Taxonomies

TAXONOMY	DESCRIPTION
ABUNDANCE_PARTICLES_TYPES	Classes of abundance of volcanic glasses in the sand and coarse silt fraction, see table 8.79 WRB 4th edition 2022
ACCUMULATION_TYPES	Types of accumulation of organic matter
ACTIVE_CACO3_CONTENT_METHODS	Laboratory methods used for active CaCO ₃ evaluation
ACTIVITIES	Geomorphic features: types of landform activities
AGGREGATE_FORMATION	Classes of soil aggregate formation after additions or after in-situ human alterations, see table 8.90 WRB 4th edition 2022
AGGREGATE_PENETRABILITY	Classes of aggregate penetrability for roots, see table 8.44 WRB 4th edition 2022
AGGREGATE_SIZES	Classes of aggregate size, see table 8.45 WRB 4th edition 2022
ALLUVIAL_TEPHRA	Types of layer composition of alluvial sediments or of tephra, see table 8.26 WRB 4th edition 2022
HUMAN_ALTERATIONS_TYPES	Types of in situ human alteration, see table 8.89 WRB 4th edition 2022
ANIMAL_ACTIVITY_TYPES	Types of animal activities, see table 8.87 WRB 4th edition 2022
ARRANGEMENT	Classes of spatial arrangement of surface cracks
ARTEFACT_SIZES	Types of artefact sizes, see table 8.37 WRB 4th edition 2022

ARTEFACT_TYPES	Types of artefact, see table 8.36 WRB 4th edition 2022
BOUNDARIES	Classes of distinctness of the layers lower boundary, see table 8.30 WRB 4th edition 2022
BULK_DENSITY_METHODS	Laboratory methods used to evaluate bulk density
CACO3_CONTENT_METHODS	Laboratory methods used to evaluate CaCO ₃
CARBONATE_CONTENTS	Classes of carbonate contents, see table 8.62 WRB 4th edition 2022
CEC_CA_MG_NA_K_METHODS	Laboratory methods used to evaluate CEC, Ca, Mg, Na, K
CEMENTATION_CLASSES	Classes of cementation in layer consistence, see table 8.73 WRB 4th edition 2022
CEMENTING_AGENTS	Classes of cementing agents in layer consistence, see table 8.35 WRB 4th edition 2022
CLASSIFICATION_SYSTEMS	Soil Classification systems
CLIMATE_KOPPEN	Classes of climate according to Köppen (1936)
COARSE_SIZES	Classes of coarse fragments size
CONSERVATION_MEASURE_TYPES	Surface - types of conservation measure
CONTINUITY_SALINE_EFFLORESCENCE_TYPES	Types of saline efflorescence on soil surface
CORINE	Classes of corine land cover
CRACKS_CONTINUITY	Classes of continuity of cracks in layers, see table 8.50 WRB 4th edition 2022

CRYOGENIC_ALTERATION	Classes of cryogenic alterations in layer permafrost, see table 8.81 WRB 4th edition 2022
CULTIVATION_TYPES	Types of cultivations
CURRENT_WEATHER	Types of current weather conditions
DEAD_PLANT_TYPES	Classes of dead residues of specific plants see table 8.93 WRB 4th edition 2022
DIRECTION_OF_PLOUGHING_TYPES	Land preparation - directions of ploughing
DISTANCE_TYPES	Classes of distances between surface cracks
DNFC_SIZES	Classes of size of secondary silica (only DN and FC), see table 8.68 WRB 4th edition 2022
DRAINAGE_CONDITIONS	Surface drainage conditions
ECOZONE_SHULTZ	Ecozones according to Schultz (2005 adapted)
EL_CONDUCTIVITY_PH_METHODS	Laboratory methods used to evaluate Electrical conductivity
EROSION_ACTIVITY	Types of surface unevenness erosion activities
EROSION_DEGREE	Types of surface unevenness erosion degree
EROSION_TYPES	Types of surface unevenness erosion
EXCHANGEABLE_CATIONS_METHODS	Laboratory methods used to evaluate exchangeable cations
FE_TOT_METHODS	Laboratory methods used to evaluate total Fe
FORM_COATINGS	Types of form of clay coating in layers, see table 14 in FAO 2006

GROUND_FORMS	Surface, types of patterned ground form
GYPSUM_CONTENT_METHODS	Laboratory methods used to evaluate gypsum content
GYPSUM_CONTENT_TYPES	Types of Gypsum content in a soil matrix (only in layers with little readily soluble salts), see table 8.66 4th edition 2022
G_CLASSIFICATION_SYSTEMS	Classification systems used for texture of fine earth
HUMAN_MADE_TYPES	Types of human-made surface unevenness
HYDRAULIC_CONDUCTIVITY_METHODS	Laboratory methods used to evaluate hydraulic conductivity
IRRIGATION_TYPES	Types of irrigation
LANDFORMS	Geomorphic features: land form types
LITHOGENIC_SIZES	Classes of sizes of lithogenic variegates, classes indicate the average length of the greatest dimension, see table 8.52 WRB 4th edition 2022
L_SLOPE_SHAPE	Layer: types of slope shape
MANNER_FAILURE	Types of manner of failure (brittleness) in layer consistence, see table 8.76 WRB 4th edition 2022
MINERAL_SHAPES	Types of mineral shapes, see table 15 FAO 2006
MINERAL_SIZES	Classes of mineral sizes, see table 15 FAO 2006
MEASURE_UNITS	Unit of measure (UoM) of values in SIS Measures
MN_ZN_CU_METHODS	Laboratory methods used to evaluate Mn, Zn, Cu

MOTTLES_BOUNDARY_TYPES	Types of layer redoximorphic mottles boundary see table 35 in FAO 2006
MOTTLES_CONTRAST	Types of layer redoximorphic mottles contrast, see table 34 in FAO 2006
NATURAL_MATERIAL_TYPES	Types of artificial additions of natural material, see table 8.88 WRB 4th edition 2022
NATURAL_SURFACE_ALTERATIONS_TYPES	Types of natural surface alterations
NATURAL_TYPES	Types of natural surface unevenness
NH4_METHODS	Laboratory methods used to evaluate NH4
NO3_METHODS	Laboratory methods used to evaluate NO3
NONMATRIX_PORES_TYPES	Types of non-matrix pores, see table 8.46 WRB 4th edition 2022
N_CONTENT_METHODS	Laboratory methods used to evaluate N content
ORGANIC_CARBON_METHODS	Laboratory methods used to evaluate organic carbon
ORGANIC_COATINGS	Types of form of organic matter coating, see table 14 FAO2006
ORGANIC_MATTER_CONTENT_METHODS	Laboratory methods used to evaluate organic matter content
ORGANIC_MINERAL_TYPES	Types of organo technic or mineral layer, see table 8.29 WRB 4th edition 2022
OXIMORPHIC_CEMENTATION	Classes of consistence of oximorphic features, see table 8.58 WRB 4th edition 2022

OXIMORPHIC_SIZES	Classes of sizes of oximorphic features, classes indicate the average length of the greatest dimension see table 8.57 WRB 4th edition 2022
PACKING_DENSITIES	Classes of layer packing density, see table 8.83 WRB 4th edition 2022
PARENT_MATERIALS	Types of parent material, see table 8.91 WRB 4th edition 2022
PAST_WEATHER	Types of past weather conditions
PATTERNED_COMPACTION_TYPES	Types of surface patterned compaction
PB_HG_CD_NI_CR_AS_CO_V_METHODS	Laboratory methods used to evaluate Pb, Hg, Cd, Ni, As, Co, V
PERMAFROST_TYPES	Types of cryogenic alterations, see table 8.81 WRB 4th edition 2022
PERSISTENCE	Classes of persistence of cracks, see table 8.49 WRB 4th edition 2022
PHOTO_TARGET_TYPES	Types of photo subjects
PLASTICITY	Type of plasticity in layer consistence
POINT_DATA_TYPES	Types of point soil data
PORE_ABUNDANCE	Classes of abundance of the non matrix pore
PORE_SIZES	Classes of sizes of the non matrix pore
POSITION	Types of profile positions

POTENZIOMETRIC_MEASURES	Potentiometric pH measurements, see table 8.71 4th edition 2022
PROD_TECHNIQUES	Special techniques to enhance site productivity in land use
PROFILE_POSITION	Types of profile positions in surface unevenness -
P_CONTENT_METHODS	Laboratory methods used to evaluate P content
REDOXIMORPHIC_LOCATIONS	Types of locations of oximorphic and reductimorphic features, see tables 8.55 and 8.56 WRB 4th edition 2022
REDOXIMORPHIC_SUBSTANCES	Types of substances of oximorphic and reductimorphic features, see tables 8.53 and 8.54 WRB 4th edition 2022
RETARDED_REACTION	Layer carbonates - types of retarded reaction with HCl, see table 8.63 WRB 4th edition 2022
RH_VALUE	Classes of Rh values, see table 8.59 WRB 4th edition 2022
RIBBONLIKE_SUBSTANCES	Classes of substances of ribbon-like accumulations, see table 8.61 WRB 4th edition 2022
ROC_TOC400_METHODS	Laboratory methods used to evaluate ROC and TOC400
ROOT_ABUNDANCE_CLASSES	Classes of abundance of roots
RRCLASSES_DRY	Classes of dry rupture resistance, see table 8.74 WRB 4th edition 2022
RRCLASSES_MOIST	Classes of moist rupture resistance, see table 8.74 WRB 4th edition 2022
SAND_CONTENT_METHODS	Laboratory methods used to evaluate sand content

STRATUM	Types of stratum in land use
STRUCTURE_GRADES	Classes of grades of structural units, see table 8.43 WRB 4th edition 2022
STRUCTURE_LEVELS	Levels of structural units
STRUCTURE_TYPES	Types of structural units, see table 8.42 WRB 4th edition 2022
SUBDIVISION_HORIZON	Types of Subdivisions of the Oa horizon, see table 8.92 WRB 4th edition 2022
SURFACE_CRACKS_PERSISTENCE	Classes of persistence of surface cracks
SURVEY_METHODS	Survey methods
SUSCEPTIBILITY	Classes of susceptibility for cementation, see table 8.75 WRB 4th edition 2022
TECHNOGENIC_DEBRIS_COVER_TYPES	Types of technogenic debris cover
TECH_ALTERATIONS	Classes of technical surface alterations
TEXTURE_CLASSES	Classes of Texture, see table 8.39 WRB 4th edition 2022
TEXTURE_SUBCLASSES	Subclasses of texture classes for sand and loamy sand, see table 8.40 WRB 4th edition 2022
THIXOTROPY_NAF	Classes of thixotropy and NaF field test, see table 8.80 WRB 4th edition 2022
TILLAGE_MODE	Types of tillage modes
USES	Land Uses

USE_OF_INPUT	Surface - Types of use of inputs
VEGETATION_TYPES	Types of not cultivated vegetation
WATER_ABOVE	Classes of presence of water
WATER_REPELLENCE	Classes of water repellence
WATER_SATURATION	Classes of water saturation see table 8.27 WRB 4th edition 2022
WATER_STATUS	Classes of soil water status see table 8.28 WRB 4th edition 2022
WEATHERING_STAGE	Types of weathering stage of size and shape, see table 8.34 WRB 4th edition 2022
WIDTH_TYPES	Classes of surface cracks width
WILTING_POINT_METHODS	Laboratory methods used in soil analysis
WIND	Classes of evidence of wind deposition, see table 8.32 WRB 4th edition 2022

3. Taxonomies Values

The API permits managing taxonomy classes .

- LIST: `/api/backoffice/taxonomy-values/`
- ELEMENT: `/api/backoffice/taxonomy-values/${Taxonomy_value}`

It is possible to request all values of a taxonomy

- `/api/backoffice/taxonomy-values/?taxonomy=${Taxonomy_name}`

FIELDS: []

The uri field, if present, represents the mapping (SameAs) to the standards.

The id field is the concatenation of the taxonomy code and the value :

- **`${taxonomy}:${value}`**

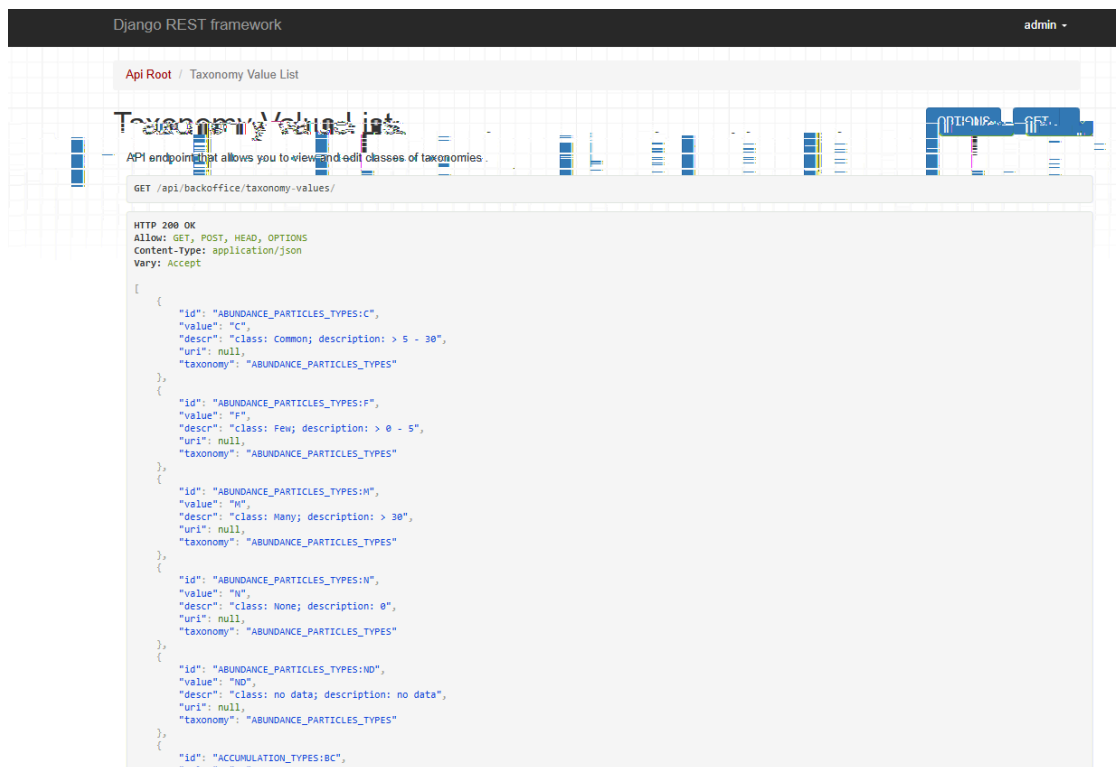


Figure 2 - MED-SIS API - Taxonomies - Classes

TAXONOMY	VALUE	DESCRIPTION
ABUNDANCE_PARTICLES_TYPES	C	class: Common - value > 5 to 30
ABUNDANCE_PARTICLES_TYPES	F	class: Few - value > 0 - 5
ABUNDANCE_PARTICLES_TYPES	M	class: Many - value > 30
ABUNDANCE_PARTICLES_TYPES	N	class: None - value == 0
ABUNDANCE_PARTICLES_TYPES	ND	No data
ACCUMULATION_TYPES	BC	Black carbon (e.g. charcoal partly charred particles soot)
ACCUMULATION_TYPES	BU	Filled earthworm burrows
ACCUMULATION_TYPES	CO	Organic matter coatings at surfaces of soil aggregates and biopore walls (no visible other material in the coatings)
ACCUMULATION_TYPES	KR	Filled krotowinas
ACCUMULATION_TYPES	ND	no data
ACCUMULATION_TYPES	NO	No visible accumulation of organic matter
ACTIVE_CACO3_CONTENT_METHODS	DM	Droineau method
ACTIVE_CACO3_CONTENT_METHODS	ND	No Data
ACTIVE_CACO3_CONTENT_METHODS	OTH	Other
ACTIVE_CACO3_CONTENT_METHODS	UN	Unknown
ACTIVITIES	A	Active or Quiescent
ACTIVITIES	I	Inactive
ACTIVITIES	ND	no data
ACTIVITIES	R	Relict
ACTIVITIES	UN	Unknown
AGGREGATE_FORMATION	N	No new granular structure present
AGGREGATE_FORMATION	ND	no data
AGGREGATE_FORMATION	P	New granular structure present in places but in other places the added or mixed materials and the previously present materials lie isolated from each other

AGGREGATE_FORMATION	T	New granular structure present throughout the layer
AGGREGATE_PENETRABILITY	N	No aggregate with dense outer rim
AGGREGATE_PENETRABILITY	ND	no data
AGGREGATE_PENETRABILITY	P	All aggregates with dense outer rim
AGGREGATE_PENETRABILITY	S	Some aggregates with dense outer rim
AGGREGATE_SIZES	CO	size: Coarse - criterion_gfp: > 5 - 10 - criterion_salpc: > 20 - 50
AGGREGATE_SIZES	EC	size: Fine - criterion_gfp: > 20 - criterion_salpc: > 100
AGGREGATE_SIZES	FI	size: Fine - criterion_gfp: > 1 - 2 - criterion_salpc: > 5 - 10
AGGREGATE_SIZES	ME	size: Medium - criterion_gfp: > 2 - 5 - criterion_salpc: > 10 - 20
AGGREGATE_SIZES	ND	size: no data - criterion_gfp: no data - criterion_salpc: no data
AGGREGATE_SIZES	VC	size: Very coarse - criterion_gfp: > 10 - 20 - criterion_salpc: > 50 - 100
AGGREGATE_SIZES	VF	size: Very fine - criterion_gfp: ≤ 1 - criterion_salpc: ≤ 5
ALLUVIAL_TEPHRA	A	Layer is composed of two or more alluvial strata
ALLUVIAL_TEPHRA	B	Layer is composed of two or more alluvial strata containing tephra
ALLUVIAL_TEPHRA	N	Layer is not composed of different strata
ALLUVIAL_TEPHRA	ND	no data
ALLUVIAL_TEPHRA	T	Layer is composed of two or more tephra strata
HUMAN_ALTERATIONS_TYPES	CP	Compaction other than a plough pan
HUMAN_ALTERATIONS_TYPES	LO	Loosening
HUMAN_ALTERATIONS_TYPES	ND	no data
HUMAN_ALTERATIONS_TYPES	NO	No in-situ alteration
HUMAN_ALTERATIONS_TYPES	OT	Other
HUMAN_ALTERATIONS_TYPES	PA	Ploughing annually
HUMAN_ALTERATIONS_TYPES	PO	Ploughing at least once every 5 years

HUMAN_ALTERATIONS_TYPES	PP	Ploughing in the past not ploughed since > 5 years
HUMAN_ALTERATIONS_TYPES	PU	Ploughing unspecified
HUMAN_ALTERATIONS_TYPES	RM	Remodelled (e.g. single ploughing)
HUMAN_ALTERATIONS_TYPES	SD	Structure deterioration other than by ploughing or remodelling
ANIMAL_ACTIVITY_TYPES	B	Bird activity
ANIMAL_ACTIVITY_TYPES	BA	Bones feathers sorted gravel of similar size
ANIMAL_ACTIVITY_TYPES	BU	Burrows (unspecified)
ANIMAL_ACTIVITY_TYPES	I	Insect activity
ANIMAL_ACTIVITY_TYPES	IA	Ant channels and nests
ANIMAL_ACTIVITY_TYPES	IO	Other insect activity
ANIMAL_ACTIVITY_TYPES	IT	Termite channels and nests
ANIMAL_ACTIVITY_TYPES	M	Mammal activity
ANIMAL_ACTIVITY_TYPES	MI	Infilled large burrows (krotovinas)
ANIMAL_ACTIVITY_TYPES	MO	Open large burrows
ANIMAL_ACTIVITY_TYPES	ND	no data
ANIMAL_ACTIVITY_TYPES	NO	No visible results of animal activity
ANIMAL_ACTIVITY_TYPES	W	Worm activity
ANIMAL_ACTIVITY_TYPES	WC	Worm casts
ANIMAL_ACTIVITY_TYPES	WE	Earthworm channels
ARRANGEMENT	I	Irreversible (persist year-round e.g. drained polder cracks cracks in cemented layers)
ARRANGEMENT	N	Non-polygonal
ARRANGEMENT	ND	No Data
ARRANGEMENT	P	Polygonal
ARRANGEMENT	R	Reversible (open and close with changing moisture e.g. in Vertisols and in soils with the Vertic or the Protovertic qualifier)

ARTEFACT_SIZES	B	size_class: Boulders - size: > 20 - 60
ARTEFACT_SIZES	C	size_class: Coarse gravel - size: > 2 - 6
ARTEFACT_SIZES	E	size_class: Fine earth - size: ≤ 0.2
ARTEFACT_SIZES	F	size_class: Fine gravel - size: > 0.2 - 0.6
ARTEFACT_SIZES	L	size_class: Large boulders - size: > 60
ARTEFACT_SIZES	M	size_class: Medium gravel - size: > 0.6 - 2
ARTEFACT_SIZES	ND	size_class: no data - size: no data
ARTEFACT_SIZES	S	size_class: Stones - size: > 6 - 20
ARTEFACT_TYPES	BA	Bottom ash
ARTEFACT_TYPES	BC	Black carbon (e.g. charcoal partly charred particles soot)
ARTEFACT_TYPES	BF	Bitumen (asphalt) fragments
ARTEFACT_TYPES	BR	Bricks adobes
ARTEFACT_TYPES	BS	Boiler slag
ARTEFACT_TYPES	BT	Bitumen (asphalt) continuous
ARTEFACT_TYPES	CE	Ceramics
ARTEFACT_TYPES	CF	Concrete fragments
ARTEFACT_TYPES	CL	Cloth carpet
ARTEFACT_TYPES	CO	Crude oil
ARTEFACT_TYPES	CR	Concrete continuous
ARTEFACT_TYPES	CU	Coal combustion byproducts
ARTEFACT_TYPES	DE	Debitage (stone tool flakes)
ARTEFACT_TYPES	DS	Dressed or crushed stones
ARTEFACT_TYPES	FA	Fly ash
ARTEFACT_TYPES	GC	Gold coins
ARTEFACT_TYPES	GF	Geomembrane fragments

ARTEFACT_TYPES	GL	Glass
ARTEFACT_TYPES	GM	Geomembrane continuous
ARTEFACT_TYPES	HW	Household waste (undifferentiated)
ARTEFACT_TYPES	IW	Industrial waste
ARTEFACT_TYPES	LL	Lumps of applied lime
ARTEFACT_TYPES	ME	Metal
ARTEFACT_TYPES	MS	Mine spoil
ARTEFACT_TYPES	ND	no data
ARTEFACT_TYPES	NO	None
ARTEFACT_TYPES	OT	Other
ARTEFACT_TYPES	OW	Organic waste
ARTEFACT_TYPES	PA	Paper cardboard
ARTEFACT_TYPES	PB	Plasterboard
ARTEFACT_TYPES	PO	Processed oil products
ARTEFACT_TYPES	PT	Plastic
ARTEFACT_TYPES	RU	Rubber (tires etc.)
ARTEFACT_TYPES	TW	Treated wood
BOUNDARIES	A	distinctness: Abrupt - mineral_organic: > 0.5 - 2 - terrestrial_organic: > 0.1 - 0.2
BOUNDARIES	C	distinctness: Clear - mineral_organic: > 2 - 5 - terrestrial_organic: > 0.2 - 0.5
BOUNDARIES	D	distinctness: Diffuse - mineral_organic: > 15 - terrestrial_organic: > 1
BOUNDARIES	G	distinctness: Gradual - mineral_organic: > 5 - 15 - terrestrial_organic: > 0.5 - 1
BOUNDARIES	U	Unknow
BOUNDARIES	V	distinctness: Very Abrupt - mineral_organic: ≤ 0.5 - terrestrial_organic: ≤ 0.1
BULK_DENSITY_METHODS	CM	Core method

BULK_DENSITY_METHODS	EM	Excavation method
BULK_DENSITY_METHODS	LM	Clod method
BULK_DENSITY_METHODS	OTH	Other method
BULK_DENSITY_METHODS	UN	Unknown method
CACO3_CONTENT_METHODS	GM	Carbonate equivalent Gravimetric method
CACO3_CONTENT_METHODS	GV	Calcimeter - Gas volumetric method
CACO3_CONTENT_METHODS	ND	No Data
CACO3_CONTENT_METHODS	OTH	Other method
CACO3_CONTENT_METHODS	TM	Titration method
CACO3_CONTENT_METHODS	UN	Unknown method
CARBONATE_CONTENTS	EX	content: Extremely calcareous- description: Extremely strong reaction thick foam forms quickly - mass: > 25
CARBONATE_CONTENTS	MO	content: Moderately calcareous- description: Visible effervescence - mass: > 2 - 10
CARBONATE_CONTENTS	NC	content: Non-calcareous- description: No visible or audible effervescence - mass: 0
CARBONATE_CONTENTS	ND	content: no data- description: no data - mass: no data
CARBONATE_CONTENTS	SL	content: Slightly calcareous- description: Audible effervescence but not visible - mass: > 0 - 2
CARBONATE_CONTENTS	ST	content: Strongly calcareous- description: Strong visible effervescence bubbles form a low foam - mass: > 10 - 25
CEC_CA_MG_NA_K_METHODS	AA	Ammonium Acetate (CH ₃ COONH ₄)
CEC_CA_MG_NA_K_METHODS	BM	BaCl ₂ /MgSO ₄ method
CEC_CA_MG_NA_K_METHODS	ME	Mehlich Extraction
CEC_CA_MG_NA_K_METHODS	ND	No Data
CEC_CA_MG_NA_K_METHODS	OTH	Other method
CEC_CA_MG_NA_K_METHODS	UN	Unknown method
CEMENTATION_CLASSES	EWC	class: Extremely weakly cemented- description: Slight force between fingers 8 - < 20 N

CEMENTATION_CLASSES	EXC	class: Extremely strongly cemented- description: Blow of ≥ 3 J (3 J = 2 kg dropped 15 cm)
CEMENTATION_CLASSES	MOC	class: Moderately cemented- description: Moderate force between hands 80 - < 160 N
CEMENTATION_CLASSES	ND	No data
CEMENTATION_CLASSES	NOC	class: Not cemented- description: Intact specimen not obtainable or very slight force between fingers < 8 N
CEMENTATION_CLASSES	STC	class: Strongly cemented- description: Foot pressure by full body weight 160 - < 800 N
CEMENTATION_CLASSES	VSC	class: Very strongly cemented- description: Blow of < 3 J (3 J = 2 kg dropped 15 cm) and does not fail under foot pressure by full body weight (800 N)
CEMENTATION_CLASSES	VWC	class: Very weakly cemented- description: Moderate force between fingers 20 - < 40 N
CEMENTATION_CLASSES	WEC	class: Weakly cemented- description: Strong force between fingers 40 - < 80 N
CEMENTING_AGENTS	CA	Secondary carbonates
CEMENTING_AGENTS	F	Fe oxides
CEMENTING_AGENTS	FH	Fe oxides in the presence of a significant concentration of organic matter
CEMENTING_AGENTS	FI	Fe oxides predominantly inside (former) soil aggregates no significant concentration of organic matter
CEMENTING_AGENTS	FN	Fe oxides no relationship to (former) soil aggregates no significant concentration of organic matter
CEMENTING_AGENTS	FO	Fe oxides predominantly on the surfaces of (former) soil aggregates no significant concentration of organic matter
CEMENTING_AGENTS	GY	Secondary gypsum
CEMENTING_AGENTS	ND	no data
CEMENTING_AGENTS	SI	Secondary silica
CLASSIFICATION_SYSTEMS	CP	class_sys: CPCS - year:NULL
CLASSIFICATION_SYSTEMS	FA	class_sys: FAO - year:NULL
CLASSIFICATION_SYSTEMS	FA67	class_sys: FAO - year:1967
CLASSIFICATION_SYSTEMS	FA74	class_sys: FAO - year:1974

CLASSIFICATION_SYSTEMS	FA85	class_sys: FAO - year:1985
CLASSIFICATION_SYSTEMS	FA88	class_sys: FAO - year:1988
CLASSIFICATION_SYSTEMS	FA94	class_sys: FAO - year:1994
CLASSIFICATION_SYSTEMS	KU	class_sys: KUBIENA - year:NULL
CLASSIFICATION_SYSTEMS	ND	No Data
CLASSIFICATION_SYSTEMS	OTH	class_sys: OTHER - year:NULL
CLASSIFICATION_SYSTEMS	RP	class_sys: Referentiel Pedologique - year:NULL
CLASSIFICATION_SYSTEMS	RP08	class_sys: Referentiel Pedologique - year:2008
CLASSIFICATION_SYSTEMS	RP95	class_sys: Referentiel Pedologique - year:1995
CLASSIFICATION_SYSTEMS	UN	class_sys: UNKNOWN - year:NULL
CLASSIFICATION_SYSTEMS	US	class_sys: USDA - year:NULL
CLASSIFICATION_SYSTEMS	US06	class_sys: USDA - year:2006
CLASSIFICATION_SYSTEMS	US10	class_sys: USDA - year:2010
CLASSIFICATION_SYSTEMS	US14	class_sys: USDA - year:2014
CLASSIFICATION_SYSTEMS	US22	class_sys: USDA - year:2022
CLASSIFICATION_SYSTEMS	US60	class_sys: USDA - year:1960
CLASSIFICATION_SYSTEMS	US75	class_sys: USDA - year:1975
CLASSIFICATION_SYSTEMS	US90	class_sys: USDA - year:1990
CLASSIFICATION_SYSTEMS	US91	class_sys: USDA - year:1991
CLASSIFICATION_SYSTEMS	US94	class_sys: USDA - year:1994
CLASSIFICATION_SYSTEMS	US98	class_sys: USDA - year:1998
CLASSIFICATION_SYSTEMS	US99	class_sys: USDA - year:1999
CLASSIFICATION_SYSTEMS	WR	class_sys: WRB - year:NULL
CLASSIFICATION_SYSTEMS	WR06	class_sys: WRB - year:2006
CLASSIFICATION_SYSTEMS	WR15	class_sys: WRB - year:2015

CLASSIFICATION_SYSTEMS	WR22	class_sys: WRB - year:2022
CLASSIFICATION_SYSTEMS	WR98	class_sys: WRB - year:1998
CLIMATE_KOPPEN	A	Tropical climates
CLIMATE_KOPPEN	AF	Tropical rainforest climate
CLIMATE_KOPPEN	AM	Tropical monsoon climate
CLIMATE_KOPPEN	AS	Tropical savanna climate with dry-summer characteristics
CLIMATE_KOPPEN	AW	Tropical savanna climate with dry-winter characteristics
CLIMATE_KOPPEN	B	Dry climates
CLIMATE_KOPPEN	BSC	Cold semi-arid climate
CLIMATE_KOPPEN	BSH	Hot semi-arid climate
CLIMATE_KOPPEN	BWC	Cold arid climate
CLIMATE_KOPPEN	BWH	Hot arid climate
CLIMATE_KOPPEN	C	Temperate climates
CLIMATE_KOPPEN	CFA	Humid subtropical climate
CLIMATE_KOPPEN	CFB	Oceanic climate
CLIMATE_KOPPEN	CFC	Subpolar oceanic climate
CLIMATE_KOPPEN	CSA	Mediterranean hot summer climate
CLIMATE_KOPPEN	CSB	Mediterranean warm/cool summer climate
CLIMATE_KOPPEN	CSC	Mediterranean cold summer climate
CLIMATE_KOPPEN	CWA	Dry-winter humid subtropical climate
CLIMATE_KOPPEN	CWB	Dry-winter subtropical highland climate
CLIMATE_KOPPEN	CWC	Dry-winter subpolar oceanic climate
CLIMATE_KOPPEN	D	Continental climates
CLIMATE_KOPPEN	DFA	Hot-summer humid continental climate
CLIMATE_KOPPEN	DFB	Warm-summer humid continental climate

CLIMATE_KOPPEN	DFC	Subarctic climate
CLIMATE_KOPPEN	DFD	Extremely cold subarctic climate
CLIMATE_KOPPEN	DSA	Mediterranean-influenced hot-summer humid continental climate
CLIMATE_KOPPEN	DSB	Mediterranean-influenced warm-summer humid continental climate
CLIMATE_KOPPEN	DSC	Mediterranean-influenced subarctic climate
CLIMATE_KOPPEN	DSD	Mediterranean-influenced extremely cold subarctic climate
CLIMATE_KOPPEN	DWA	Monsoon-influenced hot-summer humid continental climate
CLIMATE_KOPPEN	DWB	Monsoon-influenced warm-summer humid continental climate
CLIMATE_KOPPEN	DWC	Monsoon-influenced subarctic climate
CLIMATE_KOPPEN	DWD	Monsoon-influenced extremely cold subarctic climate
CLIMATE_KOPPEN	E	Polar and alpine climates
CLIMATE_KOPPEN	EF	Ice cap climate
CLIMATE_KOPPEN	ET	Tundra climate
CLIMATE_KOPPEN	ND	no data
COARSE_SIZES	B	size_cm: > 20 - 60- description: Boulders
COARSE_SIZES	C	size_cm: > 2 - 6- description: Coarse gravel
COARSE_SIZES	F	size_cm: > 0.2 - 0.6- description: Fine gravel
COARSE_SIZES	G	size_cm: > 0.2 - 6- description: Gravel
COARSE_SIZES	L	size_cm: > 60- description: Large boulders
COARSE_SIZES	M	size_cm: > 0.6 - 2- description: Medium gravel
COARSE_SIZES	N	size_cm: none- description: No coarse surface fragments
COARSE_SIZES	ND	No data
COARSE_SIZES	S	size_cm: > 6 - 20- description: Stones
CONSERVATION_MEASURE_TYPES	1	Stone line

CONSERVATION_MEASURE_TYPES	2	Bench terrace
CONSERVATION_MEASURE_TYPES	3	Graded terrace
CONSERVATION_MEASURE_TYPES	4	Contour bunds
CONSERVATION_MEASURE_TYPES	5	Graded bunch
CONSERVATION_MEASURE_TYPES	6	Drains ditches and furrows (for retention of water and/or soil)
CONSERVATION_MEASURE_TYPES	7	Grass strips (vegetated strips)
CONSERVATION_MEASURE_TYPES	8	Not applicable
CONTINUITY_SALINE_EFFLORESCENCE_TYPES	CN	Continuos
CONTINUITY_SALINE_EFFLORESCENCE_TYPES	DS	Discontinuos
CONTINUITY_SALINE_EFFLORESCENCE_TYPES	ND	No data
CORINE	1	Artificial areas
CORINE	11	Urban fabric Areas
CORINE	111	Continuous urban fabric
CORINE	112	Discontinuous urban fabric
CORINE	12	Industrial commercial and transport units
CORINE	121	Industrial or commercial units and public facilities
CORINE	122	Road and rail networks
CORINE	123	Port areas
CORINE	124	Airports
CORINE	13	Mine dump and construction sites
CORINE	131	Mineral extraction sites
CORINE	132	Dump sites.
CORINE	133	Construction sites.
CORINE	14	Artificial non-agricultural vegetated areas.
CORINE	141	Green urban areas

CORINE	142	Sport and leisure facilities
CORINE	2	Agricultural areas
CORINE	21	Arable land
CORINE	211	Non-irrigated arable land
CORINE	212	Permanently irrigated arable land
CORINE	213	Rice fields
CORINE	22	Permanent crops
CORINE	221	Vineyards
CORINE	222	Fruit tree and berry plantations
CORINE	223	Olive groves
CORINE	23	Pastures
CORINE	231	Pastures meadows and other permanent grasslands
CORINE	24	Heterogeneous agricultural areas
CORINE	241	Annual crops associated with permanent crops
CORINE	242	Complex cultivation patterns
CORINE	243	Land principally occupied by agriculture
CORINE	244	Agro-forestry areas
CORINE	3	Forest and semi-natural areas
CORINE	31	Forests
CORINE	311	Broad-leaved forest
CORINE	312	Coniferous forest
CORINE	313	Mixed forest
CORINE	32	Shrubs and/or herbaceous vegetation associations
CORINE	321	Natural grassland
CORINE	322	Moors and heathland

CORINE	323	Sclerophyllous vegetation
CORINE	324	Transitional woodland/shrub
CORINE	33	Open spaces with little or no vegetation
CORINE	331	Beaches dunes and sand plains
CORINE	332	Bare rock
CORINE	333	Sparsely vegetated areas
CORINE	334	Burnt areas
CORINE	335	Glaciers and perpetual snow
CORINE	4	Wetlands
CORINE	41	Inland wetlands
CORINE	411	Inland marshes
CORINE	412	Peat bogs
CORINE	42	Coastal wetland
CORINE	421	Coastal salt marshes
CORINE	422	Salines
CORINE	423	Intertidal flats
CORINE	5	Water bodies.
CORINE	51	Inland waters
CORINE	511	Water courses
CORINE	512	Water bodies
CORINE	52	Marine waters
CORINE	521	Coastal lagoons
CORINE	522	Estuaries
CORINE	523	Sea and ocean
CRACKS_CONTINUITY	AC	All cracks continue into the underlying layer

CRACKS_CONTINUITY	HC	At least half but not all of the cracks continue into the underlying layer
CRACKS_CONTINUITY	NC	Cracks do not continue into the underlying layer
CRACKS_CONTINUITY	ND	no data
CRACKS_CONTINUITY	SC	At least one but less than half of the cracks continue into the underlying layer
CRYOGENIC_ALTERATION	CF	Separation of coarse material and fine material
CRYOGENIC_ALTERATION	DB	Disrupted lower layer boundary
CRYOGENIC_ALTERATION	IL	Ice lens
CRYOGENIC_ALTERATION	IW	Ice wedge
CRYOGENIC_ALTERATION	MI	Mineral involutions in an organic layer
CRYOGENIC_ALTERATION	ND	no data
CRYOGENIC_ALTERATION	NO	None
CRYOGENIC_ALTERATION	OI	Organic involutions in a mineral layer
CRYOGENIC_ALTERATION	OT	Other
CULTIVATION_TYPES	ACA	Simultaneous agroforestry system with trees and annual crops
CULTIVATION_TYPES	ACB	Simultaneous agroforestry system with trees perennial and annual crops
CULTIVATION_TYPES	ACG	Simultaneous agroforestry system with trees crops and grassland
CULTIVATION_TYPES	ACP	Simultaneous agroforestry system with trees and perennial crops
CULTIVATION_TYPES	AGG	Simultaneous agroforestry system with trees and grassland
CULTIVATION_TYPES	CPA	Annual crop production (e.g. food fodder fuel fiber ornamental plants)
CULTIVATION_TYPES	CPP	Perennial crop production (e.g. food fodder fuel fiber ornamental plants)
CULTIVATION_TYPES	FDF	Fallow all plants constantly removed (dry farming)
CULTIVATION_TYPES	FOL	Fallow at least 12 months with spontaneous vegetation
CULTIVATION_TYPES	FYO	Fallow less than 12 months with spontaneous vegetation

CULTIVATION_TYPES	GIN	Intensively-managed grassland not pastured
CULTIVATION_TYPES	GIP	Intensively-managed grassland pastured
CULTIVATION_TYPES	GNP	Pasture on (semi-)natural vegetation
CULTIVATION_TYPES	ND	no data
CURRENT_WEATHER	ND	no data
CURRENT_WEATHER	OV	Overcast
CURRENT_WEATHER	PC	Partly cloudy
CURRENT_WEATHER	RA	Rain
CURRENT_WEATHER	SL	Snow
CURRENT_WEATHER	SU	Sunny/clear
CURRENT_WEATHER	WS	Sleet
DEAD_PLANT_TYPES	N	No dead plant residues
DEAD_PLANT_TYPES	ND	no data
DEAD_PLANT_TYPES	O	Other plants
DEAD_PLANT_TYPES	S	Moss fibres
DEAD_PLANT_TYPES	W	Wood
DIRECTION_OF_PLOUGHING_TYPES	0	Not applicable
DIRECTION_OF_PLOUGHING_TYPES	1	Along contours
DIRECTION_OF_PLOUGHING_TYPES	2	In slope direction tangent to contour
DISTANCE_TYPES	HU	distance_cm: > 200 - 500- description: Huge
DISTANCE_TYPES	LA	distance_cm: > 20 - 50- description: Large
DISTANCE_TYPES	ME	distance_cm: > 5 - 20- description: Medium
DISTANCE_TYPES	ND	distance_cm: no data- description: no data
DISTANCE_TYPES	SM	distance_cm: > 2 - 5- description: Small
DISTANCE_TYPES	TI	distance_cm: ≤ 0.5- description: Tiny

DISTANCE_TYPES	VH	distance_cm: > 500- description: Very huge
DISTANCE_TYPES	VL	distance_cm: > 50 -200- description: Very large
DISTANCE_TYPES	VS	distance_cm: > 0.5 - 2- description: Very small
DNFC_SIZES	CO	class: Coarse - size: > 2 - 6
DNFC_SIZES	FI	class: Fine - size: > 0.5 - 1
DNFC_SIZES	ME	class: Medium - size: > 1 - 2
DNFC_SIZES	ND	class: no data - size: no data
DNFC_SIZES	VC	class: Very coarse - size: > 6
DNFC_SIZES	VF	class: Very fine - size: ≤ 0.5
DRAINAGE_CONDITIONS	ED	Excessively drained
DRAINAGE_CONDITIONS	MWD	Moderately well drained
DRAINAGE_CONDITIONS	ND	no data
DRAINAGE_CONDITIONS	PD	Poorly drained
DRAINAGE_CONDITIONS	SA	Subaqueous
DRAINAGE_CONDITIONS	SED	Somewhat excessively drained
DRAINAGE_CONDITIONS	SPD	Somewhat poorly drained
DRAINAGE_CONDITIONS	VPD	Very poorly drained
DRAINAGE_CONDITIONS	WD	Well drained
ECOZONE_SHULTZ	BOR	Boreal zone
ECOZONE_SHULTZ	MDR	Dry mid-latitudes
ECOZONE_SHULTZ	MHU	Humid mid-latitudes
ECOZONE_SHULTZ	ND	no data
ECOZONE_SHULTZ	POS	Polar-subpolar zone
ECOZONE_SHULTZ	SWR	Subtropics with winter rain (Mediterranean climate)
ECOZONE_SHULTZ	SYR	Subtropics with year-round rain

ECOZONE_SHULTZ	TSD	Dry tropics and subtropics
ECOZONE_SHULTZ	TSR	Tropics with summer rain
ECOZONE_SHULTZ	TYR	Tropics with year-round rain
EL_CONDUCTIVITY_PH_METHODS	ND	No Data
EL_CONDUCTIVITY_PH_METHODS	OF	Soil to water 1:5
EL_CONDUCTIVITY_PH_METHODS	OO	Soil to water 1:1
EL_CONDUCTIVITY_PH_METHODS	OP	Soil to water 1:2.5
EL_CONDUCTIVITY_PH_METHODS	OT	Soil to water 1:2
EL_CONDUCTIVITY_PH_METHODS	OTH	Other method
EL_CONDUCTIVITY_PH_METHODS	SP	Saturated paste
EL_CONDUCTIVITY_PH_METHODS	UN	Unknown method
EROSION_ACTIVITY	HI	Active in historical times
EROSION_ACTIVITY	ND	no data
EROSION_ACTIVITY	NK	Period of activity not known
EROSION_ACTIVITY	PR	Active at present
EROSION_ACTIVITY	RE	Active in recent past (within the last 100 years)
EROSION_DEGREE	E	degree: Extreme- description: Substantial removal of deeper subsurface layers original ecological functions fully destroyed (badlands)
EROSION_DEGREE	M	degree: Moderate- description: Clear evidence of removal of surface layers original ecological functions partly destroyed
EROSION_DEGREE	ND	No data
EROSION_DEGREE	S	degree: Slight- description: Some evidence of damage to surface layers original ecological functions largely intact
EROSION_DEGREE	V	degree: Severe- description: Surface layers completely removed and subsurface layers exposed original ecological functions largely destroyed
EROSION_TYPES	A	Aeolian (wind) erosion
EROSION_TYPES	AO	Other types of wind erosion

EROSION_TYPES	AS	Shifting sands
EROSION_TYPES	MM	Mass movement (landslides and similar phenomena)
EROSION_TYPES	NC	Erosion not categorized
EROSION_TYPES	ND	no data
EROSION_TYPES	NO	No evidence of erosion
EROSION_TYPES	W	Water erosion
EROSION_TYPES	WA	Water and aeolian (wind) erosion
EROSION_TYPES	WG	Gully erosion
EROSION_TYPES	WR	Rill erosion
EROSION_TYPES	WS	Sheet erosion
EROSION_TYPES	WT	Tunnel erosion
EXCHANGEABLE_CATIONS_METHODS	AA	Ammonium Acetate (CH ₃ COONH ₄)
EXCHANGEABLE_CATIONS_METHODS	BM	Extraction with BaCl ₂
EXCHANGEABLE_CATIONS_METHODS	ME	Mehlich Extraction
EXCHANGEABLE_CATIONS_METHODS	ND	No Data
EXCHANGEABLE_CATIONS_METHODS	OTH	Other
EXCHANGEABLE_CATIONS_METHODS	UN	Unknown
FE_TOT_METHODS	AR	Acqua Regia

FORM_COATINGS	DC	Discontinuous circular
FORM_COATINGS	DE	Dendroidal
FORM_COATINGS	DI	Discontinuous irregular
FORM_COATINGS	ND	no data
FORM_COATINGS	O	Other
GROUND_FORMS	N	None
GROUND_FORMS	ND	no data
GROUND_FORMS	P	Polygons
GROUND_FORMS	R	Rings
GROUND_FORMS	S	Stripes
GYPSUM_CONTENT_METHODS	AT	Acetone EDTA titration (George Holmgren)
GYPSUM_CONTENT_METHODS	FA	Soluble Calcium FAAS
GYPSUM_CONTENT_METHODS	ND	No Data
GYPSUM_CONTENT_METHODS	OTH	Other method
GYPSUM_CONTENT_METHODS	UN	Unknown method
GYPSUM_CONTENT_TYPES	EX	content: Extremely calcareous - mass: > 60 - conductivity: > 1.8 dS m ⁻¹ in 10 g soil / 250 ml H ₂ O
GYPSUM_CONTENT_TYPES	MO	content: Moderately gypsiferous - mass: > 5 - 15 - conductivity: > 1.8 dS m ⁻¹ in 10 g soil / 250 ml H ₂ O
GYPSUM_CONTENT_TYPES	ND	content: no data - mass: no data - conductivity: no data
GYPSUM_CONTENT_TYPES	NG	content: Non-gypsiferous - mass: 0 - conductivity: ≤ 1.8 dS m ⁻¹ in 10 g soil / 25 ml H ₂ O or ≤ 0.18 dS m ⁻¹ in 10 g soil / 250 ml H ₂ O
GYPSUM_CONTENT_TYPES	SL	content: Slightly gypsiferous - mass: > 0 - 5 - conductivity: > 0.18 - ≤ 1.8 dS m ⁻¹ in 10 g soil / 250 ml H ₂ O
GYPSUM_CONTENT_TYPES	ST	content: Strongly gypsiferous - mass: > 15 - 60 - conductivity: > 1.8 dS m ⁻¹ in 10 g soil / 250 ml H ₂ O
G_CLASSIFICATION_SYSTEMS	OT	OTHER
G_CLASSIFICATION_SYSTEMS	U	USDA

G_CLASSIFICATION_SYSTEMS	UN	UNKNOWN
G_CLASSIFICATION_SYSTEMS	W	WRB
HUMAN_MADE_TYPES	CD	Drainage canals
HUMAN_MADE_TYPES	CI	Irrigation canals
HUMAN_MADE_TYPES	CO	Other canals
HUMAN_MADE_TYPES	EL	Other longitudinal elevations
HUMAN_MADE_TYPES	EP	Polygonal elevations
HUMAN_MADE_TYPES	ER	Rounded elevations
HUMAN_MADE_TYPES	HP	Polygonal holes
HUMAN_MADE_TYPES	HR	Rounded holes
HUMAN_MADE_TYPES	HT	Human-made terraces
HUMAN_MADE_TYPES	ND	no data
HUMAN_MADE_TYPES	NO	None
HUMAN_MADE_TYPES	OT	Other
HUMAN_MADE_TYPES	RB	Raised beds
HYDRAULIC_CONDUCTIVITY_METHODS	IN	Infiltrometer
HYDRAULIC_CONDUCTIVITY_METHODS	ND	No Data
HYDRAULIC_CONDUCTIVITY_METHODS	OTH	Other method
HYDRAULIC_CONDUCTIVITY_METHODS	UN	Unknown method
IRRIGATION_TYPES	1	Surface / gravity (can be by use borders basins furrow corrugation i.e. parallel ridges and grooves)
IRRIGATION_TYPES	2	Sprinkler (different types of sprinklers: central pivot fixed set of sprinklers travelling gun irrigation system etc.)
IRRIGATION_TYPES	3	Drip irrigation (trickle dribble or localized irrigation in which the water trickles onto or into the soil near the plant)
IRRIGATION_TYPES	4	Not applicable
LANDFORMS	A	Anthropogenic landforms

LANDFORMS	AA	Area of accumulation
LANDFORMS	AAR	Landfill of remoulded natural material
LANDFORMS	AAW	Landfill of anthropic origin waste
LANDFORMS	AC	Embankment for canal/river or other constructions
LANDFORMS	AM	Mining area
LANDFORMS	AMA	Active quarry
LANDFORMS	AMQ	Abandoned quarry
LANDFORMS	AR	Grading reshaped slope
LANDFORMS	AS	Anthropic sinkhole
LANDFORMS	AT	Terraced slope
LANDFORMS	ATD	Deteriorated terraced slope
LANDFORMS	ATI	Intact terraced slope
LANDFORMS	ATM	Mechanizable terraced slope with connected levels
LANDFORMS	E	Aeolian landform
LANDFORMS	ED	Aeolian deposit
LANDFORMS	EDA	Dune affected by anthropic influence
LANDFORMS	EDD	Dune
LANDFORMS	EDE	Dune affected by erosion
LANDFORMS	EDL	Loess
LANDFORMS	EDS	Stabilized dune
LANDFORMS	EI	Interdune area
LANDFORMS	EIL	Interdune area periodically inundated
LANDFORMS	ES	Aeolian deflation surface
LANDFORMS	G	Glacial and periglacial landforms
LANDFORMS	GA	Snow Avalanche channel

LANDFORMS	GAC	Channel
LANDFORMS	GAD	Deposit
LANDFORMS	GB	Block stream/field
LANDFORMS	GC	Glacier cirque
LANDFORMS	GD	Glacial deposit
LANDFORMS	GDA	Ablation moraine
LANDFORMS	GDD	Drumlin
LANDFORMS	GDF	Lateral/Frontal moraine
LANDFORMS	GDI	Intermoraine depression
LANDFORMS	GDS	Subglacial deposit
LANDFORMS	GF	Fluvio-glacial deposit
LANDFORMS	GFE	Esker
LANDFORMS	GFK	Kame terrace
LANDFORMS	GFS	Sandur
LANDFORMS	GH	Hummock
LANDFORMS	GN	Nivations niche
LANDFORMS	GO	Protalus ramparts
LANDFORMS	GP	Poligonal soil
LANDFORMS	GR	Rock glacier
LANDFORMS	GS	Bowl shaped depression
LANDFORMS	GSI	Infilled bowl shaped depression
LANDFORMS	GT	Glacial terrace
LANDFORMS	GU	Crioturbation unevenness
LANDFORMS	GV	Glacial valley
LANDFORMS	GVH	Hanging glacial valley

LANDFORMS	K	Karst landforms
LANDFORMS	KC	Cave
LANDFORMS	KD	Karst depression
LANDFORMS	KDD	Sinkhole (Doline)
LANDFORMS	KDP	Poje
LANDFORMS	KDR	Residual relief (hum)
LANDFORMS	KDU	Uvala
LANDFORMS	KF	Fluvio-karst valley
LANDFORMS	KFB	Blind valley
LANDFORMS	KFC	Closed valley
LANDFORMS	KFD	Dry valley
LANDFORMS	KG	Griza
LANDFORMS	KK	Karren (field) grooves
LANDFORMS	M	Marine and Lacustrine landforms
LANDFORMS	MA	Abrasion platform
LANDFORMS	MAC	Cliff foot (talus)
LANDFORMS	MC	Coastal plain
LANDFORMS	MCC	Tidal canal
LANDFORMS	MCM	Mud plain
LANDFORMS	MCR	Coastal ribbon
LANDFORMS	MCS	Sand plain
LANDFORMS	MCT	Tidal plain
LANDFORMS	MD	River delta
LANDFORMS	MDR	Reclaimed delta
LANDFORMS	ML	Lacustrine terrace

LANDFORMS	MO	Lagoon
LANDFORMS	MP	Lacustrine plain
LANDFORMS	MPM	Mineral deposit infill
LANDFORMS	MPO	Organic deposit infill (peat bog)
LANDFORMS	MR	Reclaimed draining lacustrine plain
LANDFORMS	MRM	Mineral deposit infill
LANDFORMS	MRO	Organic deposit infill (peat bog)
LANDFORMS	MT	Marine terrace
LANDFORMS	ND	no data
LANDFORMS	S	Slope landforms
LANDFORMS	SA	Deep-seated gravitational slope deformations (trenches counter slopes)
LANDFORMS	SC	Soil creep terraces
LANDFORMS	SD	Slope deposit (talus)
LANDFORMS	SF	Debris/Mud flow deposit
LANDFORMS	SG	Gelifluction lobes
LANDFORMS	SL	Translational/ Rotational Landslide
LANDFORMS	SLD	Landslide deposit
LANDFORMS	SLN	Landslide niche
LANDFORMS	SO	Soil slips deposit
LANDFORMS	SR	Rockfall and toppling
LANDFORMS	SRD	Landslide deposit
LANDFORMS	SRN	Landslide niche
LANDFORMS	SS	Solifluction lobes
LANDFORMS	ST	Talus deposit
LANDFORMS	T	Structural landforms

LANDFORMS	TC	Cuesta
LANDFORMS	TD	Structural depression (graben)
LANDFORMS	TG	Irregular slopes due to tectonic structure influence
LANDFORMS	TR	Structural relief (horst)
LANDFORMS	TS	Structural surface
LANDFORMS	TSD	Dissected structural surface
LANDFORMS	TSW	Wavy structural surface
LANDFORMS	TV	Fault slope
LANDFORMS	UN	Unknown
LANDFORMS	V	Volcanic landforms
LANDFORMS	VA	Caldera
LANDFORMS	VC	Volcanic cone
LANDFORMS	VCA	Volcanic ash cone
LANDFORMS	VCL	Volcanic lava cone
LANDFORMS	VCP	Polygenic volcanic cone
LANDFORMS	VCS	Scoria volcanic cone
LANDFORMS	VD	Lava dome
LANDFORMS	VF	Pyroclastic flow deposit
LANDFORMS	VH	Lahar deposit
LANDFORMS	VL	Lava flow
LANDFORMS	VP	Volcanic Plateau
LANDFORMS	VR	Volcanic crater
LANDFORMS	VRM	Maar crater
LANDFORMS	VT	Structural – Volcanic depression
LANDFORMS	W	Water related landforms

LANDFORMS	WA	Alluvial fan
LANDFORMS	WB	Badlands
LANDFORMS	WBB	Badlands
LANDFORMS	WBC	Calanco
LANDFORMS	WC	Glacis
LANDFORMS	WD	Colluvial deposit
LANDFORMS	WF	Flat bottom fluvial valley
LANDFORMS	WG	Gorge
LANDFORMS	WH	Hanging valley step
LANDFORMS	WL	Rills and gullies
LANDFORMS	WP	Earth pyramids
LANDFORMS	WR	Riverbed
LANDFORMS	WRA	Anastomosing
LANDFORMS	WRB	Braided
LANDFORMS	WRC	Crevasse splay
LANDFORMS	WRE	Ephemeral
LANDFORMS	WRF	Floodplain and overflow area
LANDFORMS	WRI	Fluvial island
LANDFORMS	WRM	Meandering
LANDFORMS	WRR	River bank
LANDFORMS	WRS	Single channel
LANDFORMS	WS	Spring depression
LANDFORMS	WT	Fluvial terrace
LANDFORMS	WTE	Erosion scarp
LANDFORMS	WTS	Surface

LANDFORMS	WU	Alluvial plain
LANDFORMS	WUB	Reclaimed alluvial plain
LANDFORMS	WUD	Depressed area in an alluvial plain
LANDFORMS	WV	V shaped small Valley
LANDFORMS	WW	Wide fluvial valley
LITHOGENIC_SIZES	C	class: Coarse - size: > 20
LITHOGENIC_SIZES	F	class: Fine - size: > 2 - 6
LITHOGENIC_SIZES	M	class: Medium - size: > 6 - 20
LITHOGENIC_SIZES	ND	class: no data - size: no data
LITHOGENIC_SIZES	V	class: Very fine - size: ≤ 2
L_SLOPE_SHAPE	B	shape: Broken- description: Discontinuous
L_SLOPE_SHAPE	I	shape: Irregular- description: Pockets more deep than wide
L_SLOPE_SHAPE	ND	No data
L_SLOPE_SHAPE	S	shape: Smooth- description: Nearly plane surface
L_SLOPE_SHAPE	W	shape: Wavy- description: Pockets less deep than wide
MANNER_FAILURE	BR	type: Brittle- description: Abruptly (pops or shatters)
MANNER_FAILURE	DF	type: Deformable- description: After compression to one half the original thickness
MANNER_FAILURE	ND	type: no data- description: no data
MANNER_FAILURE	SD	type: Semi-deformable- description: Before compression to one half the original thickness
MINERAL_SHAPES	A	Angular
MINERAL_SHAPES	E	Fine
MINERAL_SHAPES	F	Medium
MINERAL_SHAPES	I	Irregular
MINERAL_SHAPES	ND	no data
MINERAL_SHAPES	R	Very fine

MINERAL_SIZES	C	class: Coarse - size: > 20
MINERAL_SIZES	F	class: Fine - size: 2 - 6
MINERAL_SIZES	M	class: Medium - size: 6 - 20
MINERAL_SIZES	ND	class: no data - size: no data
MINERAL_SIZES	V	class: Very fine - size: < 2
MN_ZN_CU_METHODS	DBF	DTPA ammonium bicarbonate FAAS
MN_ZN_CU_METHODS	DBIA	DTPA ammonium bicarbonate ICP-AES
MN_ZN_CU_METHODS	DBIM	DTPA ammonium bicarbonate ICP-MS
MN_ZN_CU_METHODS	DF	DTPA FAAS
MN_ZN_CU_METHODS	DIA	DTPA ICP-AES
MN_ZN_CU_METHODS	DIM	DTPA ICP-MS
MN_ZN_CU_METHODS	EF	EDTA FAAS
MN_ZN_CU_METHODS	EIA	EDTA ICP-AES
MN_ZN_CU_METHODS	EIM	EDTA ICP-MS
MN_ZN_CU_METHODS	NAF	Nitric acid FAAS
MN_ZN_CU_METHODS	NAIA	Nitric acid ICP-AES
MN_ZN_CU_METHODS	NAIM	Nitric acid ICP-MS
MN_ZN_CU_METHODS	ND	No Data
MN_ZN_CU_METHODS	OTH	Other method
MN_ZN_CU_METHODS	UN	Unknown method
MOTTLES_BOUNDARY_TYPES	C	class: Clear - size: 0.5 - 2
MOTTLES_BOUNDARY_TYPES	D	class: Diffuse - size: > 2
MOTTLES_BOUNDARY_TYPES	ND	class: no data - size: no data
MOTTLES_BOUNDARY_TYPES	S	class: Sharp - size: < 0.5
MOTTLES_CONTRAST	D	class: Distinct: Although not striking the mottles are readily seen. The hue chroma and value of the matrix are easily distinguished from those of the mottles. They may vary by

		as much as 2.5 units of hue or several units in chroma or value
MOTTLES_CONTRAST	F	class: Faint: The mottles are evident only on close examination. Soil colours in both the matrix and mottles have closely related hues chromas and value
MOTTLES_CONTRAST	ND	No data
MOTTLES_CONTRAST	P	class: Prominent: The mottles are conspicuous and mottling is one of the outstanding features of the horizon. Hue chroma and value alone or in combination are at least several units apart
NATURAL_MATERIAL_TYPES	ML	Mineral > 2 mm
NATURAL_MATERIAL_TYPES	MS	Mineral $\leq$ 2 mm
NATURAL_MATERIAL_TYPES	ND	no data
NATURAL_MATERIAL_TYPES	NO	No additions
NATURAL_MATERIAL_TYPES	OR	Organic
NATURAL_SURFACE_ALTERATIONS_TYPES	DM	Disturbance by mammals
NATURAL_SURFACE_ALTERATIONS_TYPES	MH	Mole's heaps
NATURAL_SURFACE_ALTERATIONS_TYPES	OTH	Other specific method
NATURAL_SURFACE_ALTERATIONS_TYPES	SM	Self-mulching
NATURAL_TYPES	G	Unevenness caused by shrink-swell clays (gilgai relief)
NATURAL_TYPES	N	None
NATURAL_TYPES	ND	no data
NATURAL_TYPES	O	Other
NATURAL_TYPES	P	Unevenness caused by permafrost (palsa pingo mud boils thufurs etc.)
NH4_METHODS	K1E	KCl 1M extract
NH4_METHODS	K2E	KCl 2M extract
NH4_METHODS	ND	No Data
NH4_METHODS	OTH	Other
NH4_METHODS	UN	Unknown

NO3_METHODS	K1E	KCl 1M extract
NO3_METHODS	K2E	KCl 2M extract
NO3_METHODS	ND	No Data
NO3_METHODS	OTH	Other
NO3_METHODS	UN	Unknown
NO3_METHODS	WE	Water extract
NONMATRIX_PORES_TYPES	DT	type: Dendritic Tubular- description: Cylindrical elongated branching voids - e.g. empty root channels
NONMATRIX_PORES_TYPES	IG	type: Irregular- description: Non-connected cavities chambers - e.g. vughs - various shapes
NONMATRIX_PORES_TYPES	ND	No data
NONMATRIX_PORES_TYPES	NO	type: No non-matrix pores- description: No non-matrix pores
NONMATRIX_PORES_TYPES	OT	type: Other- description: Pores not attributable to any of the previous classes
NONMATRIX_PORES_TYPES	TU	type: Tubular- description: Cylindrical and elongated voids - e.g. worm tunnels
NONMATRIX_PORES_TYPES	VE	type: Vesicular- description: Ovoid to spherical voids - e.g. solidified pseudomorphs of entrapped gas bubbles concentrated below a crust - most common in arid and semiarid environments and in permafrost soils
N_CONTENT_METHODS	CA	Coleman nitrogen analyzer
N_CONTENT_METHODS	EA	Elemental analyzer
N_CONTENT_METHODS	KJ	Kjeldahl
N_CONTENT_METHODS	MW	Microwave
N_CONTENT_METHODS	ND	No Data
N_CONTENT_METHODS	OTH	Other method
N_CONTENT_METHODS	UN	Unknown method
ORGANIC_CARBON_METHODS	EA	Hydrometer method
ORGANIC_CARBON_METHODS	ND	No Data
ORGANIC_CARBON_METHODS	OTH	Other method

ORGANIC_CARBON_METHODS	SK	Springer & Klee
ORGANIC_CARBON_METHODS	UN	Unknown method
ORGANIC_CARBON_METHODS	WB	Sieving
ORGANIC_COATINGS	A	All sand and coarse silt grains coated without cracks
ORGANIC_COATINGS	C	Cracked coatings on sand grains
ORGANIC_COATINGS	ND	no data
ORGANIC_COATINGS	U	Uncoated sand and/or coarse silt grains
ORGANIC_MATTER_CONTENT_METHODS	MF	Muffle
ORGANIC_MATTER_CONTENT_METHODS	ND	No Data
ORGANIC_MATTER_CONTENT_METHODS	OTH	Other method
ORGANIC_MATTER_CONTENT_METHODS	UN	Unknown method
ORGANIC_MINERAL_TYPES	MI	Mineral
ORGANIC_MINERAL_TYPES	ND	no data
ORGANIC_MINERAL_TYPES	OH	Organic hydromorphic
ORGANIC_MINERAL_TYPES	OT	Organic terrestrial
ORGANIC_MINERAL_TYPES	TH	Organotechnic hydromorphic
ORGANIC_MINERAL_TYPES	TT	Organotechnic terrestrial
OXIMORPHIC_CEMENTATION	EWC	class: Extremely weakly cemented- description: Slight force between fingers 8 - < 20 N
OXIMORPHIC_CEMENTATION	MOC	class: Moderately or more cemented- description: Does not fail when applying force between fingers $\geq$ 80 N
OXIMORPHIC_CEMENTATION	NC	class: Not cemented- description: Intact specimen not obtainable or very slight force between fingers < 8 N
OXIMORPHIC_CEMENTATION	ND	No data
OXIMORPHIC_CEMENTATION	VWC	class: Very weakly cemented- description: Moderate force between fingers 20 - < 40 N
OXIMORPHIC_CEMENTATION	WEC	class: Weakly cemented- description: Strong force between fingers 40 - < 80 N
OXIMORPHIC_SIZES	CO	class: Coarse - size: > 20 - 60

OXIMORPHIC_SIZES	FI	class: Fine - size: > 2 - 6
OXIMORPHIC_SIZES	ME	class: Medium - size: > 6 - 20
OXIMORPHIC_SIZES	ND	class: no data - size: no data
OXIMORPHIC_SIZES	VC	class: Very coarse - size: > 60
OXIMORPHIC_SIZES	VF	class: Very fine - size: ≤ 2
PACKING_DENSITIES	FR	class: Firm- description: Only the knifepoint penetrates when forces are applied
PACKING_DENSITIES	IN	class: Intermediate- description: Knife penetrates half when forces are applied
PACKING_DENSITIES	LO	class: Loose- description: Knife penetrates completely when forces are applied
PACKING_DENSITIES	ND	class: no data- description: no data
PACKING_DENSITIES	VL	class: Very loose- description: Knife penetrates completely even when applying low forces
PACKING_DENSITIES	VR	class: Very firm- description: Knife does not (or only a little bit) penetrate when forces are applied
PARENT_MATERIALS	I	Igneous Rock
PARENT_MATERIALS	IF	Igneous Rock/Felsic igneous
PARENT_MATERIALS	IF1	Igneous Rock/Felsic igneous/Granite
PARENT_MATERIALS	IF2	Igneous Rock/Felsic igneous/Quartz-diorite
PARENT_MATERIALS	IF3	Igneous Rock/Felsic igneous/Grano-diorite
PARENT_MATERIALS	IF4	Igneous Rock/Felsic igneous/Diorite
PARENT_MATERIALS	IF5	Igneous Rock/Felsic igneous/Rhyolite
PARENT_MATERIALS	II	Igneous Rock/Intermediate igneous
PARENT_MATERIALS	II1	Igneous Rock/Intermediate igneous/Andesite trachyte phonolite
PARENT_MATERIALS	II2	Igneous Rock/Intermediate igneous/Diorite-syenite
PARENT_MATERIALS	IM	Igneous Rock/Mafic igneous
PARENT_MATERIALS	IM1	Igneous Rock/Mafic igneous/Gabbro

PARENT_MATERIALS	IM2	Igneous Rock/Mafic igneous/Basalt
PARENT_MATERIALS	IM3	Igneous Rock/Mafic igneous/Dolerite
PARENT_MATERIALS	IP	Igneous Rock/Pyroclastic
PARENT_MATERIALS	IP1	Igneous Rock/Pyroclastic/Tuff tuffite
PARENT_MATERIALS	IP2	Igneous Rock/Pyroclastic/Volcanic scoria/breccia
PARENT_MATERIALS	IP3	Igneous Rock/Pyroclastic/Volcanic ash
PARENT_MATERIALS	IP4	Igneous Rock/Pyroclastic/Ignimbrite
PARENT_MATERIALS	IU	Igneous Rock/Ultramafic igneous
PARENT_MATERIALS	IU1	Igneous Rock/Ultramafic igneous/Peridotite
PARENT_MATERIALS	IU2	Igneous Rock/Ultramafic igneous/Peridotite
PARENT_MATERIALS	IU3	Igneous Rock/Ultramafic igneous/Serpentinite
PARENT_MATERIALS	M	Metamorphic rock
PARENT_MATERIALS	MF	Metamorphic rock/Felsic metamorphic
PARENT_MATERIALS	MF1	Metamorphic rock/Felsic metamorphic/Quartzite
PARENT_MATERIALS	MF2	Metamorphic rock/Felsic metamorphic/Gneiss migmatite
PARENT_MATERIALS	MF3	Metamorphic rock/Felsic metamorphic/Slate phyllite (pelitic rocks)
PARENT_MATERIALS	MF4	Metamorphic rock/Felsic metamorphic/Schist
PARENT_MATERIALS	MM	Metamorphic rock/Mafic metamorphic
PARENT_MATERIALS	MM1	Metamorphic rock/Mafic metamorphic/Slate phyllite (pelitic rocks)
PARENT_MATERIALS	MM2	Metamorphic rock/Mafic metamorphic/(Green)schist
PARENT_MATERIALS	MM3	Metamorphic rock/Mafic metamorphic/Gneiss rich in Fe-Mg minerals
PARENT_MATERIALS	MM4	Metamorphic rock/Mafic metamorphic/Metamorphic limestone (marble)
PARENT_MATERIALS	MM5	Metamorphic rock/Mafic metamorphic/Amphibolite
PARENT_MATERIALS	MM6	Metamorphic rock/Mafic metamorphic/Eclogite

PARENT_MATERIALS	MU	Metamorphic rock/Ultramafic metamorphic
PARENT_MATERIALS	MU1	Metamorphic rock/Ultramafic metamorphic/Serpentinite greenstone
PARENT_MATERIALS	ND	no data
PARENT_MATERIALS	S	Sedimentary rock (consolidated)
PARENT_MATERIALS	SC	Sedimentary rock (consolidated)/Clastic sediments
PARENT_MATERIALS	SC1	Sedimentary rock (consolidated)/Clastic sediments/Conglomerate breccia
PARENT_MATERIALS	SC2	Sedimentary rock (consolidated)/Clastic sediments/Sandstone greywacke arkose
PARENT_MATERIALS	SC3	Sedimentary rock (consolidated)/Clastic sediments/Silt-mud- claystone
PARENT_MATERIALS	SC4	Sedimentary rock (consolidated)/Clastic sediments/Shale
PARENT_MATERIALS	SC5	Sedimentary rock (consolidated)/Clastic sediments/Ironstone
PARENT_MATERIALS	SE	Sedimentary rock (consolidated)/Evaporites
PARENT_MATERIALS	SE1	Sedimentary rock (consolidated)/Evaporites/Anhydrite gypsum
PARENT_MATERIALS	SE2	Sedimentary rock (consolidated)/Evaporites/Halite
PARENT_MATERIALS	SO	Sedimentary rock (consolidated)/Carbonatic organic
PARENT_MATERIALS	SO1	Sedimentary rock (consolidated)/Carbonatic organic/Limestone other carbonate rock
PARENT_MATERIALS	SO2	Sedimentary rock (consolidated)/Carbonatic organic/Marl and other mixtures
PARENT_MATERIALS	SO3	Sedimentary rock (consolidated)/Carbonatic organic/Coals bitumen and related rocks
PARENT_MATERIALS	U	Sedimentary rock (unconsolidated)
PARENT_MATERIALS	UA	Sedimentary rock (unconsolidated)/Anthropogenic technogenic
PARENT_MATERIALS	UA1	Sedimentary rock (unconsolidated)/Anthropogenic technogenic/Redeposited natural material
PARENT_MATERIALS	UA2	Sedimentary rock (unconsolidated)/Anthropogenic technogenic/Industrial/artisanal deposits

PARENT_MATERIALS	UM	Sedimentary rock (unconsolidated)/Marine estuarine
PARENT_MATERIALS	UM1	Sedimentary rock (unconsolidated)/Marine estuarine/Sand
PARENT_MATERIALS	UM2	Sedimentary rock (unconsolidated)/Marine estuarine/Clay and silt
PARENT_MATERIALS	UO	Sedimentary rock (unconsolidated)/Organic
PARENT_MATERIALS	UO1	Sedimentary rock (unconsolidated)/Organic/Rainwater-fed peat (bog)
PARENT_MATERIALS	UO2	Sedimentary rock (unconsolidated)/Organic/Groundwater-fed peat (fen)
PARENT_MATERIALS	UO3	Sedimentary rock (unconsolidated)/Organic/Lacustrine (organic limnic sediments)
PARENT_MATERIALS	UR	Sedimentary rock (unconsolidated)/Weathered residuum
PARENT_MATERIALS	UR1	Sedimentary rock (unconsolidated)/Weathered residuum/Bauxite laterite
PARENT_MATERIALS	UU	Sedimentary rock (unconsolidated)/Unspecified deposits
PARENT_MATERIALS	UU1	Sedimentary rock (unconsolidated)/Unspecified deposits/Clay
PARENT_MATERIALS	UU2	Sedimentary rock (unconsolidated)/Unspecified deposits/Loam and silt
PARENT_MATERIALS	UU3	Sedimentary rock (unconsolidated)/Unspecified deposits/Sand
PARENT_MATERIALS	UU4	Sedimentary rock (unconsolidated)/Unspecified deposits/Gravelly sand
PARENT_MATERIALS	UU5	Sedimentary rock (unconsolidated)/Unspecified deposits/Gravel broken rock
PAST_WEATHER	ND	No rain in the last 24 hours
PAST_WEATHER	NM	No rain in the last month
PAST_WEATHER	NW	No rain in the last week
PAST_WEATHER	RD	Rain but no heavy rain in the last 24 hours
PAST_WEATHER	RE	Extremely rainy or snow melting
PAST_WEATHER	RH	Heavy rain for some days or excessive rain in the last 24 hours
PATTERNED_COMPACTION_TYPES	AM	Animal paths

PATTERNED_COMPACTION_TYPES	GT	Grazing terracettes
PATTERNED_COMPACTION_TYPES	OTH	Other
PATTERNED_COMPACTION_TYPES	TP	Technogenic paths
PB_HG_CD_NI_CR_AS_CO_V_METHODS	DBF	DTPA ammonium bicarbonate FAAS
PB_HG_CD_NI_CR_AS_CO_V_METHODS	DBIA	DTPA ammonium bicarbonate ICP-AES
PB_HG_CD_NI_CR_AS_CO_V_METHODS	DBIM	DTPA ammonium bicarbonate ICP-MS
PB_HG_CD_NI_CR_AS_CO_V_METHODS	NAF	Nitric acid FAAS
PB_HG_CD_NI_CR_AS_CO_V_METHODS	NAIA	Nitric acid ICP-AES
PB_HG_CD_NI_CR_AS_CO_V_METHODS	NAIM	Nitric acid ICP-MS
PB_HG_CD_NI_CR_AS_CO_V_METHODS	ND	No Data
PB_HG_CD_NI_CR_AS_CO_V_METHODS	OTH	Other
PB_HG_CD_NI_CR_AS_CO_V_METHODS	UN	Unknown
PERMAFROST_TYPES	I	Massive ice cementation by ice or readily visible ice crystals
PERMAFROST_TYPES	N	No permafrost
PERMAFROST_TYPES	ND	no data
PERMAFROST_TYPES	T	Soil temperature of < 0 °C and insufficient water to form readily visible ice crystals
PERSISTENCE	I	Irreversible (persist year-round e.g. drained polder cracks cracks in cemented layers)
PERSISTENCE	ND	No Data
PERSISTENCE	NO	No cracks
PERSISTENCE	R	Reversible (open and close with changing moisture e.g. in Vertisols and in soils with the Vertic or the Protovertic qualifier)
PHOTO_TARGET_TYPES	pit	photo of the pit
PHOTO_TARGET_TYPES	site	photo of the site
PLASTICITY	MP	type: Moderately plastic- description: 4 mm diameter roll supports itself - 2 mm diameter roll does not
PLASTICITY	ND	type: No data- description: no data

PLASTICITY	NP	type: Non-plastic- description: Does not form a roll 6 mm in diameter or if a roll is formed it cannot support itself if held on end
PLASTICITY	SP	type: Slightly plastic- description: 6 mm diameter roll supports itself - 4 mm diameter roll does not
PLASTICITY	VP	type: Very plastic- description: 2 mm diameter roll supports itself
POINT_DATA_TYPES	A	Auger hole
POINT_DATA_TYPES	M	Minipit
POINT_DATA_TYPES	OTH	Other specific method
POINT_DATA_TYPES	P	Profile
PORE_ABUNDANCE	F	class: Few - number: > 1 - 3
PORE_ABUNDANCE	C	class: Common - number: > 3 - 5
PORE_ABUNDANCE	F	class: Fine - number: > 1 to 3
PORE_ABUNDANCE	M	class: Many - number: > 5
PORE_ABUNDANCE	ND	class: no data - number: no data
PORE_ABUNDANCE	V	class: Very Fine - number: ≤ 1
PORE_SIZES	CO	class: Coarse - area: 1 dm ² - diameter: > 5 - 10 mm
PORE_SIZES	FI	class: Fine - area: 1 cm ² - diameter: > 1 - 2 mm
PORE_SIZES	ME	class: Medium - area: 1 dm ² - diameter: > 2 - 5 mm
PORE_SIZES	ND	class: no data - area: no data - diameter: no data
PORE_SIZES	VC	class: Very Coarse - area: 1 m ² - diameter: > 10 mm
PORE_SIZES	VF	class: Very Fine - area: 1 cm ² - diameter: ≤ 1 mm
POSITION	BS	Backslope
POSITION	EB	Endorheic basin
POSITION	FS	Footslope
POSITION	ND	No Data
POSITION	NO	No uneven terrain

POSITION	OB	Basin with outflow
POSITION	SH	Shoulder
POSITION	SU	Summit
POSITION	TS	Toeslope
POSITION	VB	Valley bottom
POTENZIOMETRIC_MEASURES	C15	class: 1:5 - solution: CaCl2 0.01 M
POTENZIOMETRIC_MEASURES	K15	class: 1:5 - solution: KCl 1 M
POTENZIOMETRIC_MEASURES	ND	class: no data - solution: no data
POTENZIOMETRIC_MEASURES	W11	class: 1:1 - solution: Distilled water (H2O)
POTENZIOMETRIC_MEASURES	W15	class: 1:5 - solution: Distilled water (H2O)
PROD_TECHNIQUES	CW	Wet cultivation
PROD_TECHNIQUES	DC	Drainage by open canals
PROD_TECHNIQUES	DU	Underground drainage
PROD_TECHNIQUES	HT	Human-made terraces
PROD_TECHNIQUES	IR	Irrigation
PROD_TECHNIQUES	LO	Local raise of land surface
PROD_TECHNIQUES	ND	No data
PROD_TECHNIQUES	NO	None
PROD_TECHNIQUES	OT	Other
PROD_TECHNIQUES	RB	Raised beds
PROFILE_POSITION	E	On an unaffected surface
PROFILE_POSITION	H	On the high
PROFILE_POSITION	L	In the low
PROFILE_POSITION	ND	no data
PROFILE_POSITION	S	On the slope

P_CONTENT_METHODS	BR	Bray
P_CONTENT_METHODS	MN	Mehlich No.3
P_CONTENT_METHODS	ND	No Data
P_CONTENT_METHODS	OL	Olsen
P_CONTENT_METHODS	OTH	Other method
P_CONTENT_METHODS	UN	Unknown method
REDOXIMORPHIC_LOCATIONS	ND	No data
REDOXIMORPHIC_LOCATIONS	OI	Inner parts Inside soil aggregates
REDOXIMORPHIC_LOCATIONS	OIB	Inner parts Inside soil aggregates: both concretions and/or nodules (not possible to distinguish)
REDOXIMORPHIC_LOCATIONS	OIC	Inner parts Inside soil aggregates: concretions
REDOXIMORPHIC_LOCATIONS	OIM	Inner parts Inside soil aggregates: masses
REDOXIMORPHIC_LOCATIONS	OIN	Inner parts Inside soil aggregates: nodules
REDOXIMORPHIC_LOCATIONS	OM	Mottles
REDOXIMORPHIC_LOCATIONS	OO	Outer parts
REDOXIMORPHIC_LOCATIONS	OOA	Outer parts On surfaces of soil aggregates
REDOXIMORPHIC_LOCATIONS	OOE	Outer parts On biopore walls lining the entire wall surface
REDOXIMORPHIC_LOCATIONS	OOH	Outer parts Adjacent to surfaces of soil aggregates infused into the matrix (hypocoats)
REDOXIMORPHIC_LOCATIONS	OOI	Outer parts Adjacent to biopores infused into the matrix (hypocoats)
REDOXIMORPHIC_LOCATIONS	OON	Outer parts On biopore walls not lining the entire wall surface
REDOXIMORPHIC_LOCATIONS	OR	Random (not associated with aggregate surfaces or pores)
REDOXIMORPHIC_LOCATIONS	ORN	Random (not associated with aggregate surfaces or pores) Distributed over the layer no order visible
REDOXIMORPHIC_LOCATIONS	ORS	Random (not associated with aggregate surfaces or pores) Distributed over the layer surrounding areas with reductimorphic features
REDOXIMORPHIC_LOCATIONS	ORT	Random (not associated with aggregate surfaces or pores) Throughout

REDOXIMORPHIC_SUBSTANCES	AS	Fe and Al sulfates (not specified)
REDOXIMORPHIC_SUBSTANCES	FE	Fe oxides
REDOXIMORPHIC_SUBSTANCES	FM	Fe and Mn oxides
REDOXIMORPHIC_SUBSTANCES	FS	Fe sulfides
REDOXIMORPHIC_SUBSTANCES	JA	Jarosite
REDOXIMORPHIC_SUBSTANCES	MN	Mn oxides
REDOXIMORPHIC_SUBSTANCES	ND	No data
REDOXIMORPHIC_SUBSTANCES	NV	No visible accumulation
REDOXIMORPHIC_SUBSTANCES	SM	Schwertmannite
RETARDED_REACTION	H	Reaction with 1 M HCl only after heating
RETARDED_REACTION	I	Reaction with 1 M HCl immediate
RETARDED_REACTION	ND	No data
RH_VALUE	ND	No data
RH_VALUE	R1	processes: Methane formation - rh_value: < 10- description: Flammable methane present
RH_VALUE	R2	processes: Sulfide formation - rh_value: 10 - 13- description: Black colour due to metal sulfides (spraying with a 10% HCl solution causes the formation of H2S)
RH_VALUE	R3	processes: Formation of FeII/FeIII oxides (green rust) - rh_value: 13 - 20- description: Blue-green to grey colour Fe2+ ions always present (reduced areas show a positive alfa alfa - dipyrldyl test)
RH_VALUE	R4	processes: Redox reactions of Fe - rh_value: temporally < 20- description: Oximorphic features of Fe
RH_VALUE	R5	processes: Redox reactions of Mn - rh_value: temporally 20 - 29- description: Oximorphic features of Mn - temporally no free oxygen present
RH_VALUE	R6A	processes: Strongly aerated - rh_value: > 33- description: No redoximorphic features
RH_VALUE	R6D	processes: Denitrification - rh_value: 29 - 33- description: No redoximorphic features
RIBBONLIKE_SUBSTANCES	CC	Clay minerals

RIBBONLIKE_SUBSTANCES	CH	Clay minerals and organic matter
RIBBONLIKE_SUBSTANCES	CO	Clay minerals and Fe oxides and/or Mn oxides
RIBBONLIKE_SUBSTANCES	HH	Organic matter
RIBBONLIKE_SUBSTANCES	ND	No data
RIBBONLIKE_SUBSTANCES	NO	No ribbon-like accumulations
RIBBONLIKE_SUBSTANCES	OH	Fe oxides and/or Mn oxides and organic matter
RIBBONLIKE_SUBSTANCES	OO	Fe oxides and/or Mn oxides
RIBBONLIKE_SUBSTANCES	TO	Clay minerals Fe oxides and/or Mn oxides and organic matter
ROC_TOC400_METHODS	LEC	LECO RC612 analyser
ROC_TOC400_METHODS	ND	No Data
ROC_TOC400_METHODS	OTH	Other
ROC_TOC400_METHODS	UN	Unknown
ROOT_ABUNDANCE_CLASSES	A	class: Abundant - number<=2mm: > 50 - number<0.5mm: > 50 - number0.5to2mm: > 50 - number>2mm: > 20 - number2to5mm: > 20 - number>5mm: > 20
ROOT_ABUNDANCE_CLASSES	C	class: Common - number<=2mm: 11 - 20 - number<0.5mm: 11 - 20 - number0.5to2mm: 11 - 20 - number>2mm: 6 - 10 - number2to5mm: 6 - 10 - number>5mm: 6 - 10
ROOT_ABUNDANCE_CLASSES	F	class: Few - number<=2mm: 6 - 10 - number<0.5mm: 6 - 10 - number0.5to2mm: 6 - 10 - number>2mm: 3 - 5 - number2to5mm: 3 - 5 - number>5mm: 3 - 5
ROOT_ABUNDANCE_CLASSES	M	class: Many - number<=2mm: 21 - 50 - number<0.5mm: 21 - 50 - number0.5to2mm: 21 - 50 - number>2mm: 11 - 20 - number2to5mm: 11 - 20 - number>5mm: 11 - 20
ROOT_ABUNDANCE_CLASSES	N	class: None - number<=2mm: 0 - number<0.5mm: 0 - number0.5to2mm: 0 - number>2mm: 0 - number2to5mm: 0 - number>5mm: 0
ROOT_ABUNDANCE_CLASSES	ND	No data
ROOT_ABUNDANCE_CLASSES	V	class: Very few - number<=2mm: 1 - 5 - number<0.5mm: 1 - 5 - number0.5to2mm: 1 - 5 - number>2mm: 1 - 2 - number2to5mm: 1 - 2 - number>5mm: 1 - 2
RRCLASSES_DRY	EH	class: Extremely hard- description: Foot pressure by full body weight [160 .. 800[N

RRCLASSES_DRY	HA	class: Hard- description: Strong force between fingers [40 .. 80[N
RRCLASSES_DRY	LO	class: Loose- description: Intact specimen not obtainable
RRCLASSES_DRY	MH	class: Moderately hard- description: Moderate force between fingers [20 .. 40[N
RRCLASSES_DRY	ND	No data
RRCLASSES_DRY	RI	class: Rigid- description: Blow of < 3 J (3 J = 2 kg dropped 15 cm) and does not fail under foot pressure by full body weight (800 N)
RRCLASSES_DRY	SH	class: Slightly hard- description: Slight force between fingers [8 .. 20[N
RRCLASSES_DRY	SO	class: Soft- description: Very slight force between fingers < 8 N
RRCLASSES_DRY	VH	class: Very hard- description: Moderate force between hands [80 .. 160[N
RRCLASSES_DRY	VR	class: Very rigid- description: Blow of ≥ 3 J (3 J = 2 kg dropped 15 cm)
RRCLASSES_MOIST	EI	class: Extremely firm- description: Moderate force between hands 80 - < 160 N
RRCLASSES_MOIST	FI	class: Firm- description: Moderate force between fingers 20 - < 40 N
RRCLASSES_MOIST	FR	class: Friable- description: Slight force between fingers 8 - < 20
RRCLASSES_MOIST	LO	class: Loose- description: Intact specimen not obtainable
RRCLASSES_MOIST	ND	No data
RRCLASSES_MOIST	RI	class: Rigid- description: Blow of < 3 J (3 J = 2 kg dropped 15 cm) and does not fail under foot pressure by full body weight (800 N)
RRCLASSES_MOIST	SR	class: Slightly rigid- description: Foot pressure by full body weight 160 - < 800 N
RRCLASSES_MOIST	VF	class: Very friable- description: Very slight force between fingers < 8 N
RRCLASSES_MOIST	VI	class: Very firm- description: Strong force between fingers 40 - < 80 N
RRCLASSES_MOIST	VR	class: Very rigid- description: Blow of ≥ 3 J (3 J = 2 kg dropped 15 cm)

SAND_CONTENT_METHODS	BH	Hydrometer method
SAND_CONTENT_METHODS	LD	Laser diffraction
SAND_CONTENT_METHODS	OTH	Other method
SAND_CONTENT_METHODS	PR	Pipette and gravimetric method
SAND_CONTENT_METHODS	SV	Sieving
SAND_CONTENT_METHODS	UN	Unknown method
SB_METHODS	ND	No Data
SB_METHODS	OTH	Other
SB_METHODS	UN	Unknown
SEALING_AGENT	BA	Biological by algae
SEALING_AGENT	BC	Biological by cyanobacteria
SEALING_AGENT	BF	Biological by fungi
SEALING_AGENT	BL	Biological by lichens
SEALING_AGENT	BM	Biological by mosses
SEALING_AGENT	CC	Chemical by carbonates
SEALING_AGENT	CG	Chemical by gypsum
SEALING_AGENT	CR	Chemical by readily soluble salts
SEALING_AGENT	CS	Chemical by silica
SEALING_AGENT	ND	No data
SEALING_AGENT	NO	No crust present
SEALING_AGENT	PD	Physical only when dry
SEALING_AGENT	PP	Physical permanent
SEASON	AU	Autumn
SEASON	DS	Dry season
SEASON	ND	No data

SEASON	NS	No significant seasonality for plant growth
SEASON	SP	Spring
SEASON	SU	Summer
SEASON	WI	Winter
SEASON	WS	Wet season
SECONDARY_CARBONATE_TYPES	AS	Coatings on soil aggregate surfaces or biopore walls
SECONDARY_CARBONATE_TYPES	FI	Filaments (including continuous filaments like pseudomycelia)
SECONDARY_CARBONATE_TYPES	MA	Masses (including spheroidal aggregations like white eyes (byeloglaska))
SECONDARY_CARBONATE_TYPES	NC	Nodules and/or concretions
SECONDARY_CARBONATE_TYPES	ND	No data
SECONDARY_CARBONATE_TYPES	NO	No secondary carbonates
SECONDARY_CARBONATE_TYPES	UR	Coatings on undersides of coarse fragments and of remnants of broken-up cemented layers
SECONDARY_SILICA_TYPES	CH	Accumulations within a layer cemented by secondary silica
SECONDARY_SILICA_TYPES	DN	Nodules (durinodes)
SECONDARY_SILICA_TYPES	FC	Remnants of a layer that has been cemented by secondary silica
SECONDARY_SILICA_TYPES	ND	No data
SECONDARY_SILICA_TYPES	NO	No secondary silica
SECONDARY_SILICA_TYPES	OT	Other accumulations
GYP SUM_TYPES	CR	Gypsum crystal intergrowths or nodules
GYP SUM_TYPES	DP	Disperse powdery gypsum
GYP SUM_TYPES	FL	Filaments (vermiform gypsum pseudomycelia)
GYP SUM_TYPES	ND	No data
GYP SUM_TYPES	OT	Other
GYP SUM_TYPES	SC	Soft concretions

SHAPES	CC	vertical curvature Concave horizontal curvature Concave
SHAPES	CL	vertical curvature Concave horizontal curvature Linear
SHAPES	CV	vertical curvature Concave horizontal curvature Convex
SHAPES	LC	vertical curvature Linear horizontal curvature Concave
SHAPES	LL	vertical curvature Linear horizontal curvature Linear
SHAPES	LV	vertical curvature Linear horizontal curvature Convex
SHAPES	ND	No Data
SHAPES	NO	no slope
SHAPES	VC	vertical curvature Convex horizontal curvature Concave
SHAPES	VL	vertical curvature Convex horizontal curvature Linear
SHAPES	VV	vertical curvature Convex horizontal curvature Convex
SHAPE_TYPES	B	shape: Broken- description: Discontinuous
SHAPE_TYPES	I	shape: Irregular- description: Pockets more deep than wide
SHAPE_TYPES	ND	shape: no data- description: no data
SHAPE_TYPES	S	shape: Smooth- description: Nearly plane surface
SHAPE_TYPES	W	shape: Wavy- description: Pockets less deep than wide
SIGN_OF_PLOUGHING_TYPES	0	No visible signs of tillage
SIGN_OF_PLOUGHING_TYPES	1	No-till
SIGN_OF_PLOUGHING_TYPES	2	Signs of tillage in the past: ridges visible forming a pattern but ridges less pronounced/flattened
SIGN_OF_PLOUGHING_TYPES	3	Signs of field recently tilled: patterns of ridges clearly visible that form a pattern
SILT_CLAY_CONTENT_METHODS	BH	Hydrometer method
SILT_CLAY_CONTENT_METHODS	LD	Laser diffraction
SILT_CLAY_CONTENT_METHODS	OTH	Other method
SILT_CLAY_CONTENT_METHODS	PR	Pipette and gravimetric method
SILT_CLAY_CONTENT_METHODS	UN	Unknown method

SIZE_SHAPE_TYPES	B	size: > 20 - 60 size_class: Boulders
SIZE_SHAPE_TYPES	BA	Boulders - Angular
SIZE_SHAPE_TYPES	BB	Boulders - Rounded and angular
SIZE_SHAPE_TYPES	BF	Boulders - Flat
SIZE_SHAPE_TYPES	BR	Boulders - Rounded
SIZE_SHAPE_TYPES	BS	Boulders - Subrounded
SIZE_SHAPE_TYPES	C	size: > 2 - 6 size_class: Coarse gravel
SIZE_SHAPE_TYPES	CA	Coarse gravel - Angular
SIZE_SHAPE_TYPES	CB	Coarse gravel - Rounded and angular
SIZE_SHAPE_TYPES	CF	Coarse gravel - Flat
SIZE_SHAPE_TYPES	CR	Coarse gravel - Rounded
SIZE_SHAPE_TYPES	CS	Coarse gravel - Subrounded
SIZE_SHAPE_TYPES	F	size: > 0.2 - 0.6 size_class: Fine gravel
SIZE_SHAPE_TYPES	FA	Fine gravel - Angular
SIZE_SHAPE_TYPES	FB	Fine gravel- Rounded and angular
SIZE_SHAPE_TYPES	FF	Fine gravel - Flat
SIZE_SHAPE_TYPES	FR	Fine gravel - Rounded
SIZE_SHAPE_TYPES	FS	Fine gravel - Subrounded
SIZE_SHAPE_TYPES	G	size: > 0.2 - 6 size_class: Gravel
SIZE_SHAPE_TYPES	GA	Gravel - Angular
SIZE_SHAPE_TYPES	GB	Gravel - Rounded and angular
SIZE_SHAPE_TYPES	GF	Gravel - Flat
SIZE_SHAPE_TYPES	GR	Gravel - Rounded
SIZE_SHAPE_TYPES	GS	Gravel - Subrounded
SIZE_SHAPE_TYPES	L	size: > 60 size_class: Large boulders

SIZE_SHAPE_TYPES	LA	Large boulders - Angular
SIZE_SHAPE_TYPES	LB	Large boulders - Rounded and angular
SIZE_SHAPE_TYPES	LF	Large boulders - Flat
SIZE_SHAPE_TYPES	LR	Large boulders - Rounded
SIZE_SHAPE_TYPES	LS	Large boulders - Subrounded
SIZE_SHAPE_TYPES	M	size: > 0.2 - 0.6 size_class: Medium gravel
SIZE_SHAPE_TYPES	MA	Medium gravel - Angular
SIZE_SHAPE_TYPES	MB	Medium gravel - Rounded and angular
SIZE_SHAPE_TYPES	MF	Medium gravel - Flat
SIZE_SHAPE_TYPES	MR	Medium gravel - Rounded
SIZE_SHAPE_TYPES	MS	Medium gravel - Subrounded
SIZE_SHAPE_TYPES	ND	no data
SIZE_SHAPE_TYPES	NO	None
SIZE_SHAPE_TYPES	S	size: > 6 - 20 size_class: Stones
SIZE_SHAPE_TYPES	SA	Stones - Angular
SIZE_SHAPE_TYPES	SB	Stones - Rounded and angular
SIZE_SHAPE_TYPES	SF	Stones - Flat
SIZE_SHAPE_TYPES	SR	Stones - Rounded
SIZE_SHAPE_TYPES	SS	Stones - Subrounded
SOIL_MOIST_REGIME	AQ	Udic
SOIL_MOIST_REGIME	AR	Aridic
SOIL_MOIST_REGIME	ND	Udic
SOIL_MOIST_REGIME	PA	Peraquic
SOIL_MOIST_REGIME	PU	Perudic
SOIL_MOIST_REGIME	TO	Torric

SOIL_MOIST_REGIME	UD	Udic
SOIL_MOIST_REGIME	US	Ustic
SOIL_MOIST_REGIME	XE	Xeric
SOIL_TEMP_REGIME	CR	Cryic
SOIL_TEMP_REGIME	FR	Frigid
SOIL_TEMP_REGIME	HT	Hyperthermic
SOIL_TEMP_REGIME	IF	Isofrigid
SOIL_TEMP_REGIME	IH	Isohyperthermic
SOIL_TEMP_REGIME	IM	Isomesic
SOIL_TEMP_REGIME	IT	Isothermic
SOIL_TEMP_REGIME	ME	Mesic
SOIL_TEMP_REGIME	ND	No data
SOIL_TEMP_REGIME	PE	Pergelic
SOIL_TEMP_REGIME	TH	Thermic
SOLUBLE_CATIONS_METHODS	SSE	Saturated soil extract
SOLUBLE_CATIONS_METHODS	ND	No Data
SOLUBLE_CATIONS_METHODS	OTH	Other
SOLUBLE_CATIONS_METHODS	UN	Unknown
STICKINESS	ND	type: no data- description: no data
STICKINESS	NST	type: Non-sticky- description: After release of pressure practically no soil material adheres to thumb and finger
STICKINESS	SST	type: Slightly sticky- description: After pressure soil material adheres to both thumb and finger but comes off one or the other rather cleanly. It is not appreciably stretched when the digit are separated
STICKINESS	ST	type: Sticky- description: After pressure soil material adheres to both thumb and finger and tends to stretch somewhat and pull apart rather than pulling free from either digit
STICKINESS	VST	type: Very sticky- description: After pressure soil material

		adheres strongly to both thumb and finger and is decidedly stretched when they are separated
STRATUM	GS	Ground stratum
STRATUM	MS	Mid-stratum
STRATUM	US	Upper stratum
STRUCTURE_GRADES	M	grade: Moderate- description: The units are well formed and evident in place. When disturbed, the soil material parts into a mixture of mostly whole units, some broken units, and material that is not in units. Aggregates part from adjoining aggregates to reveal nearly entire faces that have properties distinct from those of fractured surfaces.
STRUCTURE_GRADES	ND	No data
STRUCTURE_GRADES	S	grade: Strong- description: The units are distinct in place. When disturbed, they separate cleanly, mainly into whole units. Aggregates have distinct surface properties.
STRUCTURE_GRADES	W	grade: Weak- description: The units are barely observable in place. When gently disturbed, the soil material parts into a mixture of whole and broken units, the majority of which exhibit no surfaces of weakness. The surfaces differ in some way from the interiors.
STRUCTURE_LEVELS	1	Second Level Structure Type 1
STRUCTURE_LEVELS	10	Third Level Structure Type 1.1.1
STRUCTURE_LEVELS	11	Third Level Structure Type 1.2.1
STRUCTURE_LEVELS	12	Third Level Structure Type 2.1.1
STRUCTURE_LEVELS	13	Third Level Structure Type 2.2.1
STRUCTURE_LEVELS	14	Third Level Structure Type 3.1.1
STRUCTURE_LEVELS	15	Third Level Structure Type 3.2.1
STRUCTURE_LEVELS	2	Second Level Structure Type 2
STRUCTURE_LEVELS	3	Second Level Structure Type 3
STRUCTURE_LEVELS	4	Second Level Structure Type 1.1
STRUCTURE_LEVELS	5	Second Level Structure Type 1.2
STRUCTURE_LEVELS	6	Second Level Structure Type 2.1
STRUCTURE_LEVELS	7	Second Level Structure Type 2.2

STRUCTURE_LEVELS	8	Second Level Structure Type 3.1
STRUCTURE_LEVELS	9	Second Level Structure Type 3.2
STRUCTURE_TYPES	BA	type: Angular blocky - formation: Soil aggregate structure natural
STRUCTURE_TYPES	BS	type: Subangular blocky - formation: Soil aggregate structure natural
STRUCTURE_TYPES	CL	type: Cloddy - formation: Artificial structural elements
STRUCTURE_TYPES	CO	type: Columnar - formation: Soil aggregate structure natural
STRUCTURE_TYPES	FE	type: Flat-edged - formation: Soil aggregate structure natural
STRUCTURE_TYPES	GR	type: Granular - formation: Soil aggregate structure natural
STRUCTURE_TYPES	LC	type: Lenticular - formation: Soil aggregate structure natural
STRUCTURE_TYPES	M	type: Massive - No structural units
STRUCTURE_TYPES	MR	type: Massive - formation: No structural units rock structure inherited from the parent material structure not changing with soil moisture not or only slightly chemically weathered
STRUCTURE_TYPES	MS	type: Massive - formation: No structural units soil structure present when moist and changing into soil aggregate structure when dry
STRUCTURE_TYPES	MW	type: Massive - formation: No structural units rock structure inherited from the parent material structure not changing with soil moisture strongly chemically weathered (e.g. saprolite)
STRUCTURE_TYPES	ND	type: no data - formation: no data
STRUCTURE_TYPES	PH	type: Polyhedral - formation: Soil aggregate structure natural
STRUCTURE_TYPES	PL	type: Platy - formation: Soil aggregate structure natural or resulting from artificial pressure
STRUCTURE_TYPES	PR	type: Prismatic - formation: Soil aggregate structure natural
STRUCTURE_TYPES	PS	type: Pseudosand/Pseudosilt - formation: Soil aggregate structure natural
STRUCTURE_TYPES	S	type: Single grain - No structural units
STRUCTURE_TYPES	SR	type: Single grain - formation: No structural units rock structure inherited from the parent material

STRUCTURE_TYPES	SS	type: Single grain - formation: No structural units soil structure resulting from soil-forming processes like loss of organic matter and/or oxides and/or clay minerals or loss of stratification
STRUCTURE_TYPES	ST	type: Stratified - formation: No structural units rock structure visible stratification from sedimentation
STRUCTURE_TYPES	WE	type: Wedge-shaped - formation: Soil aggregate structure natural
SUBDIVISION_HORIZON	CO	type: Compact- description: Breaks into longitudinal pieces with unsharp edges
SUBDIVISION_HORIZON	CR	type: Crumbly- description: Breaks into crumbly pieces or breaks powdery
SUBDIVISION_HORIZON	ND	No data
SUBDIVISION_HORIZON	SE	type: Sharp-edged- description: Breaks into longitudinal pieces with sharp edges
SURFACE_CRACKS_PERSISTENCE	I	Irreversible (persist year-round, e.g., drained polder cracks, cracks in cemented layers)
SURFACE_CRACKS_PERSISTENCE	ND	No data
SURFACE_CRACKS_PERSISTENCE	R	Reversible (open and close with changing moisture, e.g., in Vertisols and in soils with the Vertic or the Protovertic qualifier)
SURVEY_METHODS	CP	method: CPCS - year:NULL
SURVEY_METHODS	FA	method: FAO - year:NULL
SURVEY_METHODS	FA06	method: FAO - year: 2006
SURVEY_METHODS	FA52	method: FAO - year: 1952
SURVEY_METHODS	FA77	method: FAO - year: 1977
SURVEY_METHODS	FA90	method: FAO - year: 1990
SURVEY_METHODS	ND	No data
SURVEY_METHODS	OTH	method: OTHER - year:NULL
SURVEY_METHODS	UN	method: UNKNOWN - year:NULL
SURVEY_METHODS	US	method: USDA soil survey manual - year:NULL
SURVEY_METHODS	US18	method: USDA soil survey manual - year: 2018

SURVEY_METHODS	US52	method: USDA soil survey manual - year: 1952
SURVEY_METHODS	US62	method: USDA soil survey manual - year: 1962
SURVEY_METHODS	US75	method: USDA soil survey manual - year: 1975
SURVEY_METHODS	US93	method: USDA soil survey manual - year: 1993
SURVEY_METHODS	WR	method: WRB - year:NULL
SURVEY_METHODS	WR22	method: WRB - year: 2022
SUSCEPTIBILITY	CW	Cementation after repeated drying and wetting
SUSCEPTIBILITY	ND	no data
SUSCEPTIBILITY	NO	No cementation after repeated drying and wetting
TECHNOGENIC_DEBRIS_COVER_TYPES	AN	Artesanal natural material
TECHNOGENIC_DEBRIS_COVER_TYPES	ID	Industrial dust
TECHNOGENIC_DEBRIS_COVER_TYPES	MM	Mixed material
TECHNOGENIC_DEBRIS_COVER_TYPES	OG	Organic garbage
TECHNOGENIC_DEBRIS_COVER_TYPES	PS	Pavements and paving stones
TECHNOGENIC_DEBRIS_COVER_TYPES	SL	Synthetic liquid
TECHNOGENIC_DEBRIS_COVER_TYPES	SS	Synthetic solid
TECHNOGENIC_DEBRIS_COVER_TYPES	WL	Waste liquid
TECH_ALTERATIONS	LV	Levelling
TECH_ALTERATIONS	ND	no data
TECH_ALTERATIONS	NO	None
TECH_ALTERATIONS	OT	Other
TECH_ALTERATIONS	SA	Sealing by asphalt
TECH_ALTERATIONS	SC	Sealing by concrete
TECH_ALTERATIONS	SO	Other types of sealing
TECH_ALTERATIONS	TR	Topsoil removal

TEXTURE_CLASSES	C	class: Clay sand:≤ 45 silt: < 40 clay:≥ 40
TEXTURE_CLASSES	CL	class: Clay loam sand:> 20 to ≤ 45 silt: > 15 to < 53 clay:≥ 27 to < 40
TEXTURE_CLASSES	L	class: Loam sand:> 23 to ≤ 52 silt: ≥ 28 to < 50 clay:≥ 7 to < 27
TEXTURE_CLASSES	LS	class: Loamy sand sand:> 70 to ≤ 90 silt: < 30 clay:< 15
TEXTURE_CLASSES	S	class: Sand sand:> 85 silt: < 15 clay:< 10
TEXTURE_CLASSES	SC	class: Sandy clay sand:> 45 to ≤ 65 silt: < 20 clay:≥ 35 to < 55
TEXTURE_CLASSES	SCL	class: Sandy clay loam sand:> 45 to ≤ 80 silt: < 28 clay:≥ 20 to < 35
TEXTURE_CLASSES	SI	class: Silt sand:≤ 20 silt: ≥ 80 clay:< 12
TEXTURE_CLASSES	SIC	class: Silty clay sand:≤ 20 silt: ≥ 40 to ≤ 60 clay:≥ 40 to ≤ 60
TEXTURE_CLASSES	SICL	class: Silty clay loam sand:≤ 20 silt: > 40 to ≤ 73 clay:≥ 27 to < 40
TEXTURE_CLASSES	SIL	class: Silt loam sand:≤ 50 -- ≥ 8 silt: ≥ 50 to < 80 -- ≥ 80 to < 88 clay:< 27 -- ≥ 12 to ≤ 20
TEXTURE_CLASSES	SL	class: Sandy loam sand:> 52 to ≤ 85 -- > 43 to ≤ 52 silt: ≤ 48 -- ≥ 41 to < 50 clay:< 20 -- < 7
TEXTURE_SUBCLASSES	CS	subclass: Coarse sand feel:Grainy very_coarse_sand: ≥ 25 medium_sand:< 50 sum: Not defined
TEXTURE_SUBCLASSES	FS	subclass: Fine sand feel:Grainy very_coarse_sand: Not defined medium_sand:Not defined sum: Not defined -- < 25
TEXTURE_SUBCLASSES	LCS	subclass: Loamy coarse sand feel:Grainy very_coarse_sand: ≥ 25 medium_sand:< 50 sum: Not defined
TEXTURE_SUBCLASSES	LFS	subclass: Loamy fine sand feel:Grainy very_coarse_sand: Not defined medium_sand:Not defined sum: Not defined -- < 25
TEXTURE_SUBCLASSES	LMS	subclass: Loamy medium sand feel:Grainy very_coarse_sand: < 25 -- ≥ 25 medium_sand:Not defined -- ≥ 50 sum: ≥ 25 -- Not define
TEXTURE_SUBCLASSES	LVFS	subclass: Loamy very fine sand feel:Tending to be floury very_coarse_sand: Not defined medium_sand:Not defined sum: Not defined
TEXTURE_SUBCLASSES	MS	subclass: Medium sand feel:Grainy very_coarse_sand: < 25 -- ≥ 25 medium_sand:Not defined -- ≥ 50 sum: ≥ 25 -- Not defined

TEXTURE_SUBCLASSES	ND	subclass: no data feel:no data very_coarse_sand: no data medium_sand:no data sum: no data
TEXTURE_SUBCLASSES	VFS	subclass: Very fine sand feel:Tending to be floury very_coarse_sand: Not defined medium_sand:Not defined sum: Not defined
THIXOTROPY_NAF	ND	no data
THIXOTROPY_NAF	NF	Positive NaF test
THIXOTROPY_NAF	NO	None of the above
THIXOTROPY_NAF	NT	Positive NaF test and thixotropy
THIXOTROPY_NAF	TH	Thixotropy
TILLAGE_MODE	0	Not applicable
TILLAGE_MODE	1	Manual (hoe)
TILLAGE_MODE	2	Animal traction (buffalo bullock cow horse mule/donkey)
TILLAGE_MODE	3	Mechanical (tractor)
USES	A	Crop agriculture (cropping)
USES	AA	Annual field cropping
USES	AA1	Fallow system cultivation
USES	AA2	Ley system cultivation
USES	AA3	Rainfed arable cultivation
USES	AA4	Shifting cultivation
USES	AA5	Wet rice cultivation
USES	AA6	Irrigated cultivation
USES	AP	Perennial field cropping
USES	AP1	Non-irrigated cultivation
USES	AP2	Irrigated cultivation
USES	AT	Tree and shrub cropping
USES	AT1	Non-irrigated tree crop cultivation

USES	AT2	Irrigated tree crop cultivation
USES	AT3	Non-irrigated shrub crop cultivation
USES	AT4	Irrigated shrub crop cultivation
USES	F	Forestry
USES	FN	Natural forest and woodland
USES	FN1	Selective felling
USES	FN2	Clear felling
USES	FP	Plantation forestry
USES	H	Animal husbandry
USES	HE	Extensive grazing
USES	HE1	Nomadism
USES	HE2	Semi-nomadism
USES	HE3	Ranching
USES	HI	Intensive grazing
USES	HI1	Animal production
USES	HI2	Dairyng
USES	M	Mixed farming
USES	MF	Agroforestry
USES	MP	Agropastoralism
USES	ND	no data
USES	O	Other land uses
USES	P	Natural protection
USES	PD	Degradation control
USES	PD1	Without interface
USES	PD2	With interface

USES	PN	Nature and game preservation
USES	PN1	Reserves
USES	PN2	Parks
USES	PN3	Wildfire management
USES	S	Settlement Industry
USES	SC	Recreational Use
USES	SD	Disposal sites
USES	SI	Industrial Use
USES	SR	Residential Use
USES	ST	Transport
USES	SX	Excavations
USES	U	Not used and not managed
USES	Y	Military area
USE_OF_INPUT	0	No visible signs
USE_OF_INPUT	1	Crop residues
USE_OF_INPUT	2	Green manures and/or compost
USE_OF_INPUT	3	Manure (FYM cattle manure chicken manure etc.)
USE_OF_INPUT	4	Inorganic fertilizer
VEGETATION_TYPES	AF	life_form: Aquatic - Algae: fresh or brackish
VEGETATION_TYPES	AH	life_form: Aquatic - Higher aquatic plants (woody or non-woody)
VEGETATION_TYPES	AM	life_form: Aquatic - Algae: marine
VEGETATION_TYPES	CR	life_form: Surface crusts - Biological crust (of cyanobacteria algae fungi lichens and/or mosses)
VEGETATION_TYPES	ND	no data
VEGETATION_TYPES	NF	life_form: Terrestrial non-woody plants - Fungi
VEGETATION_TYPES	NG	life_form: Terrestrial non-woody plants - Grasses and/or

		herbs
VEGETATION_TYPES	NL	life_form: Terrestrial non-woody plants - Lichens
VEGETATION_TYPES	NM	life_form: Terrestrial non-woody plants - Mosses (non-peat)
VEGETATION_TYPES	NO	life_form: None (barren)- Water rock or soil surface with < 0.5% vegetation cover
VEGETATION_TYPES	NP	life_form: Terrestrial non-woody plants - Peat
VEGETATION_TYPES	WE	life_form: Terrestrial woody plants - Evergreen trees (mainly not planted)
VEGETATION_TYPES	WG	life_form: Terrestrial woody plants - Evergreen shrubs
VEGETATION_TYPES	WH	life_form: Terrestrial woody plants - Heath or dwarf shrubs
VEGETATION_TYPES	WP	life_form: Terrestrial woody plants - Plantation forest not in rotation with cropland or grassland
VEGETATION_TYPES	WR	life_form: Terrestrial woody plants - Plantation forest in rotation with cropland or grassland
VEGETATION_TYPES	WS	life_form: Terrestrial woody plants - Seasonally green shrubs
VEGETATION_TYPES	WT	life_form: Terrestrial woody plants - Seasonally green trees (mainly not planted)
WATER_ABOVE	FF	Submerged by remote flowing inland water at least once a year
WATER_ABOVE	FO	Submerged by remote flowing inland water less than once a year
WATER_ABOVE	FP	Permanently submerged by inland water
WATER_ABOVE	GF	Submerged by rising local groundwater at least once a year
WATER_ABOVE	GO	Submerged by rising local groundwater less than once a year
WATER_ABOVE	MO	Occasional storm surges (above mean high water springs)
WATER_ABOVE	MP	Permanently submerged by seawater (below mean low water springs)
WATER_ABOVE	MT	Tidal area (between mean low and mean high water springs)
WATER_ABOVE	ND	No Data
WATER_ABOVE	NO	None of the above

WATER_ABOVE	RF	Submerged by local rainwater at least once a year
WATER_ABOVE	RO	Submerged by local rainwater less than once a year
WATER_ABOVE	UF	Submerged by inland water of unknown origin at least once a year
WATER_ABOVE	UO	Submerged by inland water of unknown origin less than once a year
WATER_REPELLENCE	N	Water infiltrates completely within < 60 seconds
WATER_REPELLENCE	R	Water stands for ≥ 60 seconds
WATER_SATURATION	GF	Saturated by groundwater or flowing water for ≥ 30 consecutive days with water that has an electrical conductivity of < 4 dS m ⁻¹
WATER_SATURATION	GS	Saturated by groundwater or flowing water for ≥ 30 consecutive days with water that has an electrical conductivity of ≥ 4 dS m ⁻¹
WATER_SATURATION	MI	Saturated by water from melted ice for ≥ 30 consecutive days
WATER_SATURATION	MS	Saturated by seawater for ≥ 30 consecutive days
WATER_SATURATION	MT	Saturated by seawater according to tidal changes
WATER_SATURATION	ND	no data
WATER_SATURATION	NO	None of the above
WATER_SATURATION	PW	Pure water covered by floating organic material
WATER_SATURATION	RA	Saturated by rainwater for ≥ 30 consecutive days
WATER_STATUS	DR	moisture_class: Dry - moistening: Going dark - crushing: Makes no dust
WATER_STATUS	MO	moisture_class: Moist - moistening: No change of colour - crushing: Makes no dust
WATER_STATUS	ND	moisture_class: no data - moistening: no data - crushing: no data
WATER_STATUS	SM	moisture_class: Slightly moist - moistening: Going slightly dark - crushing: Makes no dust
WATER_STATUS	VD	moisture_class: Very dry - moistening: Going very dark - crushing: Dusty or hard
WATER_STATUS	WE	moisture_class: Wet - moistening: No change of colour - crushing: Drops of water

WEATHERING_STAGE	F	weathering_stage: Fresh - No or little signs of weathering
WEATHERING_STAGE	M	weathering_stage: Moderately weathered - Loss of original rock colour and loss of crystal form in the outer parts - centres remain relatively fresh - original strength relatively well preserved
WEATHERING_STAGE	ND	weathering_stage: no data - no data
WEATHERING_STAGE	S	weathering_stage: Strongly weathered - All but the most resistant minerals weathered - original rock colour lost throughout - tend to disintegrate under only moderate pressure
WIDTH_TYPES	FI	width_cm: > 1 - 2 - Fine
WIDTH_TYPES	ME	width_cm: > 2 - 5 - Medium
WIDTH_TYPES	ND	no data
WIDTH_TYPES	NO	No surface cracks
WIDTH_TYPES	VF	width_cm: ≤ 1 - Very fine
WIDTH_TYPES	VW	width_cm: > 10 - Very wide
WIDTH_TYPES	WI	width_cm: > 5 - 10 - Wide
WILTING_POINT_METHODS	ND	No Data
WILTING_POINT_METHODS	OTH	Other method
WILTING_POINT_METHODS	PP	Pressure Plate Method
WILTING_POINT_METHODS	UN	Unknown method
WIND	CB	Aeroturbation (cross-bedding)
WIND	ND	no data
WIND	NO	No evidence of wind deposition
WIND	OT	Other
WIND	RC	≥ 10% of the particles of medium sand or coarser are rounded or subangular and have a matt surface but only in in-blown material that has filled cracks
WIND	RH	≥ 10% of the particles of medium sand or coarser are rounded or subangular and have a matt surface
MEASURE_UNITS	mg/kg	mg/kg

MEASURE_UNITS	Mg.ha ⁻¹	Mg.ha ⁻¹
MEASURE_UNITS	cmol/kg	cmol/kg
MEASURE_UNITS	g/kg	g/kg
MEASURE_UNITS	mm/h	mm/h
MEASURE_UNITS	kg.m ⁻²	kg.m ⁻²
MEASURE_UNITS	dS/m	dS/m
MEASURE_UNITS	Mg.h.MJ ⁻¹ .m m ⁻¹	Mg.h.MJ ⁻¹ .mm ⁻¹
MEASURE_UNITS	cmol(c) ^{^(1/2)} .L ^{^(1/2)}	cmol(c) ^{^(1/2)} .L ^{^(1/2)}
MEASURE_UNITS	Unitless	Unitless
MEASURE_UNITS	g/cm ³	g/cm ³
MEASURE_UNITS	percentage	percentage
MEASURE_UNITS	cm	cm